


# HTMLDOC 1.8.15 Software Users Manual

ESP−003−20011026

Easy Software Products

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# Introduction

This document describes how to use the *HTMLDOC* software, version 1.8.15. *HTMLDOC* converts Hyper−Text Markup Language ("HTML") input files into indexed HTML, Adobe® PostScript®, or Adobe Portable Document Format ("PDF") files.

*HTMLDOC* supports most HTML 3.2 elements, some HTML 4.0 elements, and can generate title and table of contents pages. It does not currently support stylesheets.

*HTMLDOC* can be used as a standalone application, in a batch document processing environment, or as a web−based report generation application.

No restrictions are placed upon the output produced by *HTMLDOC*.

## History

Like many programs *HTMLDOC* was developed in response to a need our company had for generating high−quality documentation in printed and electronic forms. For a while we used FrameMaker® and a package from `sgi` that generated "compiled" Standard Generalized Markup Language ("SGML") files that could be used by the Electronic Book Technologies ("EBT") documentation products (EBT is now owned by INSO.) When `sgi` stopped supporting these tools we turned to INSO, but the cost of their tools is prohibitive to small businesses.

In the end we decided to write our own program to generate our documentation. HTML seemed to be the source format of choice since WYSIWYG HTML editors are widely (and freely) available and at worst you can use a plain text editor. We needed HTML output for documentation on our web server, PDF for customers

to read and/or print from their computers, and PostScript for our own printing needs.

The result of our efforts is the *HTMLDOC* software which is available for UNIX® and Microsoft® Windows®. Among other things, this software users manual is produced using *HTMLDOC*.

## Organization of This Manual

This manual is organized into tutorial and reference chapter:


- Chapter 1 − Installing HTMLDOC
- Chapter 2 − Getting Started
- Chapter 3 − Generating Books
- Chapter 4 − HTMLDOC from the Command−Line
- Chapter 5 − HTMLDOC from a Web Server
- Chapter 6 − HTML Reference
- Chapter 7 − GUI Reference
- Chapter 8 − Command−Line Reference
- Appendix A − GNU General Public License
- Appendix B − Book File Format
- Appendix C − Release Notes


## Support

Commercial support is available from Easy Software Products. Information can be found at the *HTMLDOC* web page, "http://www.easysw.com/htmldoc".

## Encryption Support

*HTMLDOC* includes code to encrypt PDF document files using the RC4 algorithm with up to a 128−bit key. While this software and code may be freely used and exported under current US laws, other countries may restrict your use and possession of this code and software.

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A copy of the GNU General Public License is included in Appendix A of this manual. If this appendix is missing from your copy of HTMLDOC, write to the Free Software Foundation, Inc., 59 Temple Place − Suite 330, Boston, MA 02111−1307, USA.

This software is based in part on the work of the Independent JPEG Group.

# Chapter 1 − Installing HTMLDOC

This chapter describes the steps needed to install *HTMLDOC* on your system from any of the source or binary distributions.

## Installing a Binary Distribution

*HTMLDOC* binary distributions are available for a number of UNIX and Windows platforms.

### Requirements

*HTMLDOC* requires approximately 2MB of disk space and one of the following environments:

- AIX 4.3.3 or higher
- Compaq Tru64 UNIX 4.0 or higher
- HP−UX 10.20 or higher
- IRIX 5.3 or higher
- Linux 2.0 or higher
- Microsoft Windows 95/98/Me
- Microsoft Windows NT 4.0
- Microsoft Windows 2000
- Solaris 2.5 or higher

## Installing HTMLDOC under Debian GNU/Linux

Run the following command to install *HTMLDOC* under Debian GNU/Linux:

```
% dselect install htmldoc-version-linux-2.0-intel.deb ENTER
```

## Uninstalling HTMLDOC under Debian GNU/Linux

Run the following command to remove *HTMLDOC* under Debian GNU/Linux:

```
% dselect remove htmldoc ENTER
```

## Installing HTMLDOC under Red Hat Linux

Run the following command to install *HTMLDOC* under Red Hat Linux:

```
% rpm -i htmldoc-version-linux-2.0-intel.rpm ENTER
```

## Uninstalling HTMLDOC under Red Hat Linux

Run the following command to remove *HTMLDOC* from your Red Hat Linux system:

```
% rpm -e htmldoc ENTER
```

## Installing HTMLDOC under UNIX

Run the following commands to install *HTMLDOC* under UNIX:

```
% gunzip htmldoc-version-platform.tar.gz ENTER
% tar xf htmldoc-version-platform.tar ENTER
% ./setup ENTER
```

Substitute the correct version and platform strings as appropriate.

## Uninstalling HTMLDOC under UNIX

Run the following command to remove *HTMLDOC* from your UNIX system:

```
% /etc/software/htmldoc.remove ENTER
```

## Installing HTMLDOC under Windows

*HTMLDOC* is provided in a self−extracting installation file under Windows. Double−click on the setup icon to install HTMLDOC under Windows.

## Uninstalling HTMLDOC under Windows

Open the Control Panel window and double−click on the *Add/Remove Software* icon. When the available software list is displayed, select HTMLDOC and click on the *Remove* button.

# Installing HTMLDOC from the Source Distribution

The complete source to *HTMLDOC* is available to build HTMLDOC for different directories, architectures, or operating systems.

## Requirements

*HTMLDOC* requires ANSI C and C++ compilers − recent versions of GCC/EGCS work fine. To build the GUI you'll also need:

- Fast Light Tool Kit ("FLTK"), version 1.0 or newer (version 1.0.11 preferred).
- X11 libraries, R5 or higher (needed to build under UNIX and OS/2 only.)

Secure (https) URL support can be enabled via the OpenSSL library. You should use at least version 0.9.6.

## Configuring the UNIX Source

*HTMLDOC* uses a configuration script produced by GNU autoconf to configure itself for your system. If your ANSI C compiler is not called *cc* or *gcc*, set the `CC` environment variable to the name and path of your ANSI C compiler:

```
% setenv CC /path/to/compiler ENTER          [C Shell]
% CC=/path/to/compiler; export CC ENTER      [Bourne/Korn Shell]
```

Similarly, if your C++ compiler is not called *CC*, *gcc*, *c++*, or *g++*, set the `CXX` environment variable to the name and path of your C++ compiler:

```
% setenv CXX /path/to/compiler ENTER         [C Shell]
% CXX=/path/to/compiler; export CXX ENTER    [Bourne/Korn Shell]
```

Then run the following command to configure *HTMLDOC* for installation in the default directories:

```
% ./configure ENTER
```

The default configuration will install *HTMLDOC* in the */usr/bin* directory with the data files under */usr/share/htmldoc* and the documentation and on−line help under */usr/share/doc/htmldoc*. Use the `--prefix` option to change the installation prefix to */usr/local*:

```
% ./configure --prefix=/usr/local ENTER
```

If the FLTK library is not installed in a standard location for your compilers, use the `--with-fltk-includes` and `--with-fltk-libs` options to point to the FLTK library:

```
% ./configure --with-fltk-libs=/path/to/fltk/lib \
        --with-fltk-includes=/path/to/fltk ENTER
```

Finally, if the OpenSSL library is not installed in a standard location for your compilers, use the `--with-openssl-includes` and `--with-openssl-libs` options to point to the OpenSSL library:

```
% ./configure --with-openssl-libs=/path/to/openssl/lib \
        --with-openssl-includes=/path/to/openssl ENTER
```

## Compiling under UNIX

*HTMLDOC* is built from a Makefile in the distribution's main directory. Simply run the "make" command to build *HTMLDOC*:

```
% make ENTER
```

If you get any fatal errors, please subscribe to the HTMLDOC mailing list and send a copy of the make/compiler output to "htmldoc@easysw.com" for assistance. Please note the version of *HTMLDOC* that you are using as well as any pertinent system information (operating system, OS version, compiler, etc.)

To subscribe to the HTMLDOC mailing list, send a message to "majordomo@easysw.com" with the text:

```
subscribe htmldoc
```

in the message body. *You must subscribe to the list to post questions and comments.*

## Installing under UNIX

To install *HTMLDOC* simply run the "make install" command:

```
% make install ENTER
```

If you are installing in a restricted directory like */usr* then you'll need to be logged in as root.

## Compiling with Visual C++

A Visual C++ 6.0 workspace file and associated project files are included in the source distribution under the "visualc" directory. Open the workspace file "htmldoc.dsw", adjust the FLTK include and project file locations, and then build the HTMLDOC target.

## Installing with Visual C++

The Windows installation package is created using InstallShield for Visual C++ 6. The "visualc/HTMLDOC" directory contains the installation information for *HTMLDOC* needed to build a binary distribution with InstallShield.

To install *HTMLDOC* without InstallShield, create an installation directory and copy the *htmldoc.exe* executable, the *afm* directory, the *data* directory, and the *doc* directory to it.

Then use the *regedit* program to create the following two string entries:

*HKEY_LOCAL_MACHINE\Software\Easy Software Products\HTMLDOC\data*
    C:\installation\directory
*HKEY_LOCAL_MACHINE\Software\Easy Software Products\HTMLDOC\doc*
    C:\installation\directory\doc

# Chapter 2 − Getting Started

This chapter describes how to start *HTMLDOC* and convert HTML files into PostScript and PDF files.

## Starting HTMLDOC

To start *HTMLDOC* under UNIX type:

```
% htmldoc ENTER
```

Choose *HTMLDOC* from the *Start* menu to start *HTMLDOC* under Windows.

## Choosing a HTML File

The *HTMLDOC* window (Figure 2−1) shows the list of input files that will be converted. Start by clicking on the *Web Page* radio button (1) to specify that you will be converting a HTML web page file.

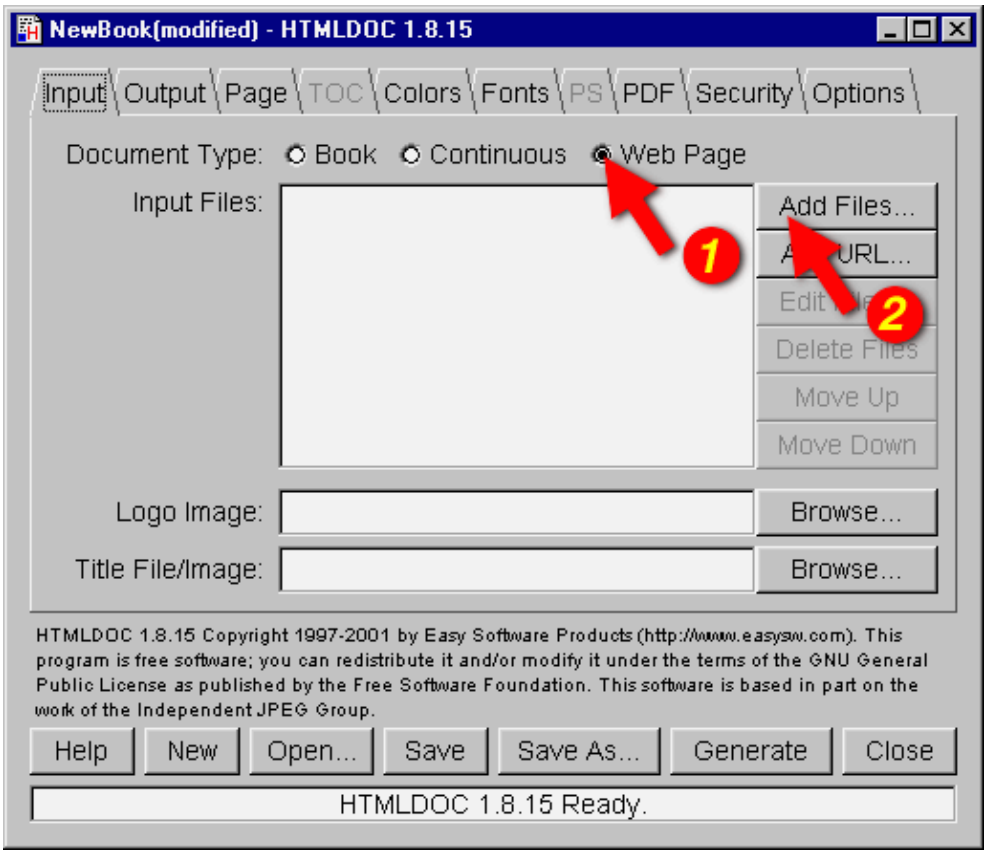



Figure 2−1 − The HTMLDOC Window

Then choose a file for conversion by clicking on the *Add Files...* button (2). When the file chooser dialog appears (Figure 3), double−click on the HTML file (3) you wish to convert from the list of files.

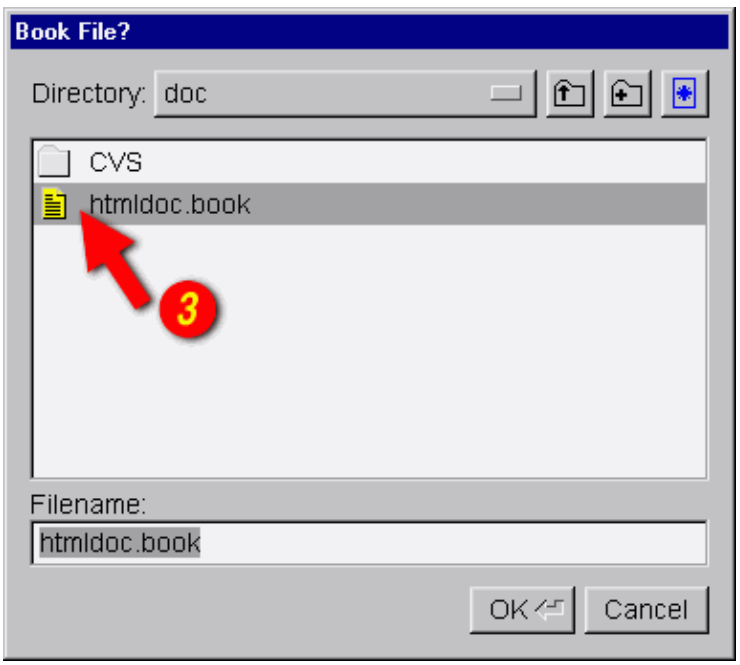



Figure 2−2 − The File Chooser Dialog

## Setting the Output File

Now that you've chosen a HTML file to convert, click on the *Output* tab (4) to set the output file (Figure 2−3). Type the name of the output file into the *Output Path* field or click on the *Browse...* button (5) to select the output file using the file chooser.

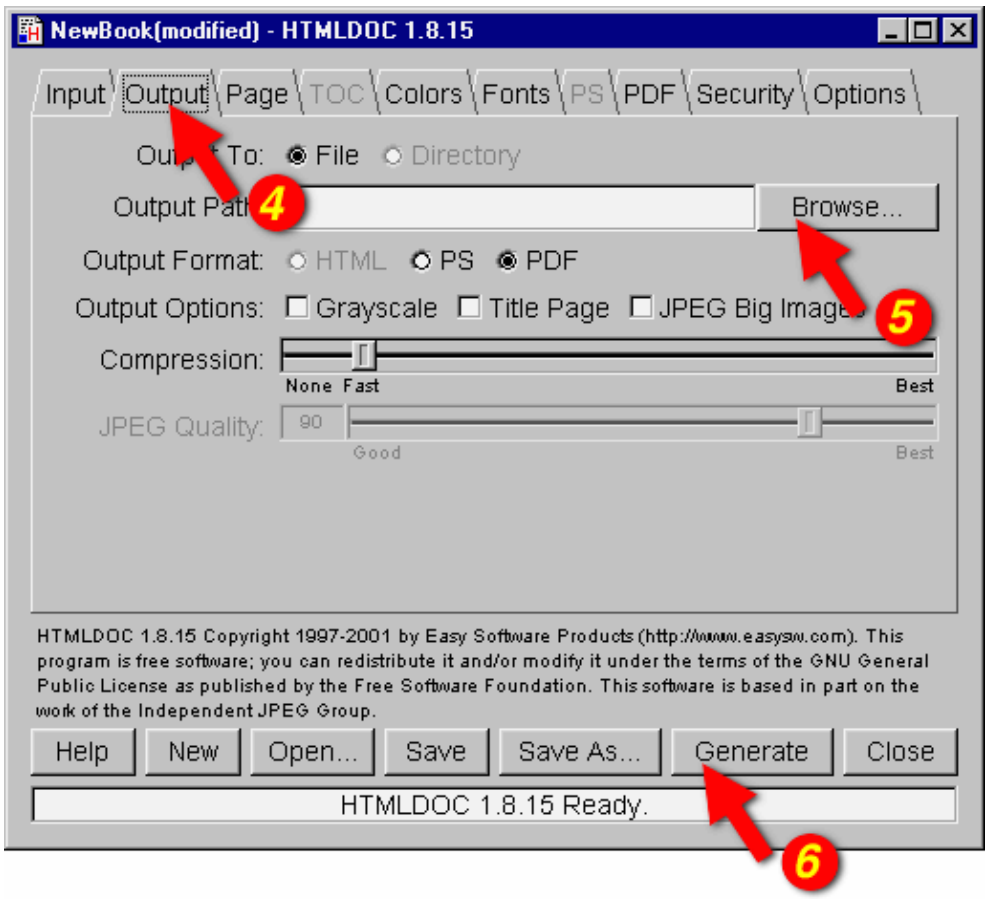



Figure 2−3 − The Output Tab

Since you chose to convert a *Web Page* instead of a book, *HTMLDOC* has automatically chosen to produce a PDF file.

## Generating the Document

Once you have chosen the output file you can generate it by clicking on the *Generate* button (6) at the bottom of the *HTMLDOC* window. When the conversion is completed you can open the PDF file that is produced using Adobe Acrobat Reader or any other PDF viewing application.

# Chapter 3 − Generating Books

This chapter describes how to generate whole books from HTML files.

## Overview

While *HTMLDOC* can convert web pages into PostScript and PDF files, its real strength is generating indexed HTML, PostScript, or PDF books.

*HTMLDOC* uses HTML heading elements to delineate chapters and headings in a book. The `H1` element is used for chapters:

```
<HTML>
<HEAD>
        <TITLE>The Little Computer that Could</TITLE>
</HEAD>
<BODY>
<H1>Chapter 1 − The Little Computer is Born</H1>
...
<H1>Chapter 2 − Little Computer's First Task</H1>
...
</BODY>
</HTML>
```

Sub−headings are marked using the `H2` through `H6` elements.

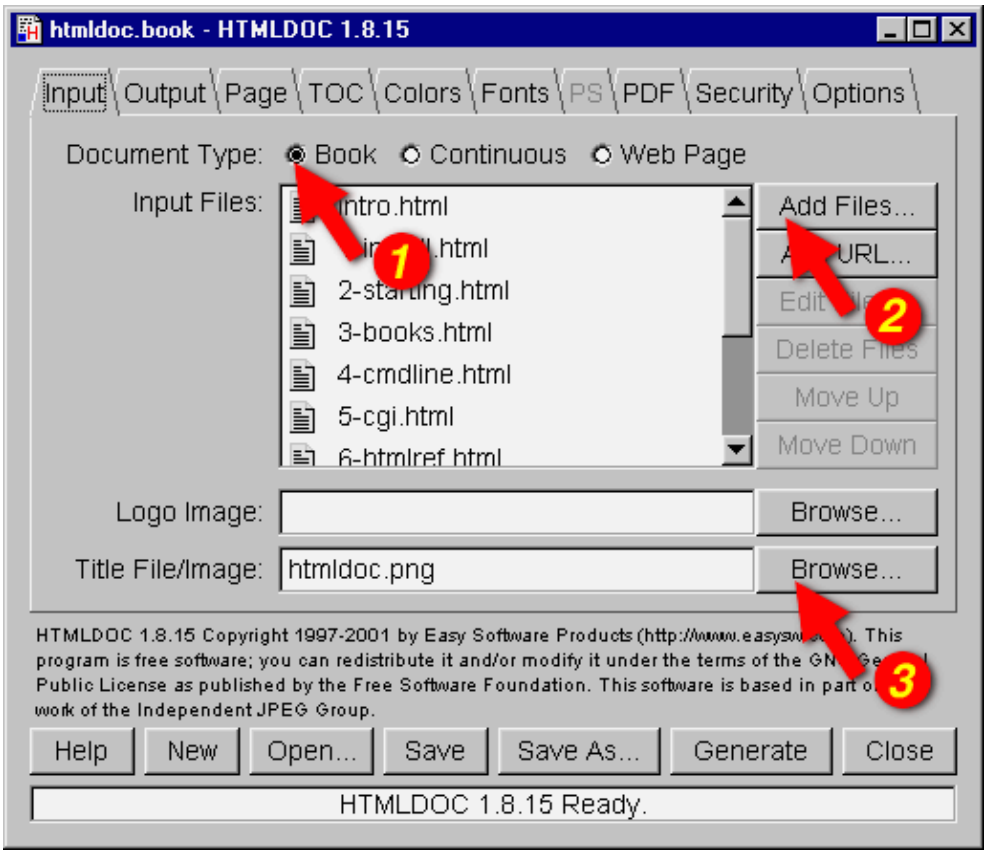



Figure 3−1: The Input Tab

## Choosing HTML Files

Start by clicking on the *Book* radio button (1) to specify you'll be converting one or more HTML files into a book.

Then choose one or more files for conversion by clicking on the *Add Files...* button (2). When the file chooser dialog appears, pick the file(s) you wish to convert from the list of files and then click on the *OK* button.

## Selecting a Title File

*HTMLDOC* supports automatic generation of a title page using an image file, the title text, and other `META` information on it. Type the title image filename into the *Title File* field or click on the *Browse...* button (3) to select a title image for your book.

*HTMLDOC* can also use a HTML file that you have generated for the title page(s). To use a HTML title page, type the title filename into the *Title File* field or click on the *Browse...* button (3) to select a HTML file for your book.

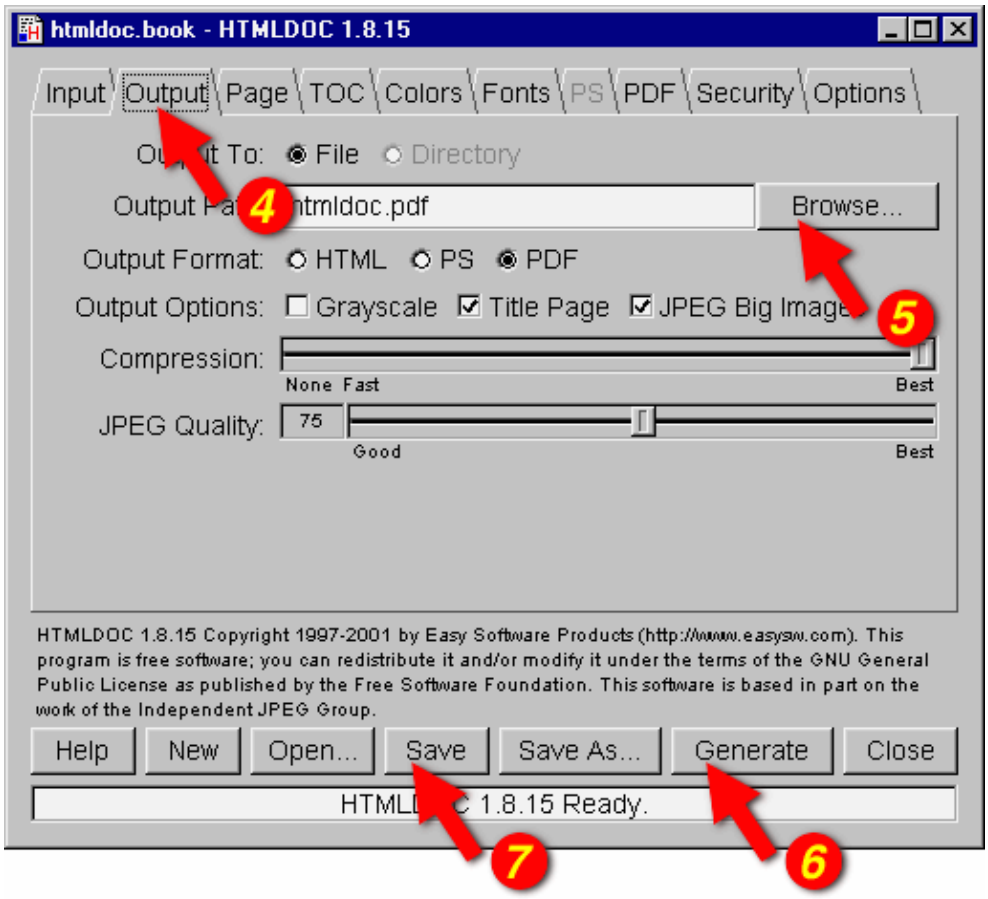



Figure 3−2: The Output Tab

## Setting the Output Format

The output format is set in the *Output* tab (4). Click on the *Output* tab and then click on the *HTML*, *PS*, or *PDF* radio buttons to set the output format.

## Setting the Output File

Now that you've chosen an output format, type the name of the output file into the *Output Path* field or click on the *Browse...* button (5) to select the output file using the file chooser.

## Generating the Document

Once you have chosen the output file you can generate it by clicking on the *Generate* button (6) at the bottom of the *HTMLDOC* window.

## Saving Your Book

*HTMLDOC* can save the list of HTML files, the title file, and all other options to a special `.BOOK` file so you can regenerate your book when you make changes to your HTML files.

Click on the *Save* button (7) to save the current book to a file.

# Chapter 4 − HTMLDOC from the Command−Line

This chapter describes how to use *HTMLDOC* from the command−line to convert web pages and generate books.

> **Note:** The free version of *HTMLDOC* for Windows does not include the command−line program.

## Converting Web Pages

To convert a web page type:

```
% htmldoc --webpage -f output.pdf filename.html ENTER
% htmldoc --webpage -f output.ps filename.html ENTER
```

where `output.pdf` and `output.ps` are the names of the files you want to generate, and `filename.html` is the HTML file you are converting.

The `--webpage` option tells *HTMLDOC* that you want to convert a web page or other unstructured document.

The `-f` option tells *HTMLDOC* what file to generate. If you don't specify an output file then a PDF file is send to the standard output.

# Generating Books

To generate a book from one or more HTML files type:

```
% htmldoc --book -f output.html file1.html ... fileN.html ENTER
% htmldoc --book -f output.pdf file1.html ... fileN.html ENTER
% htmldoc --book -f output.ps file1.html ... fileN.html ENTER
```

where `output.html`, `output.pdf`, and `output.ps` are the names of the files you want to generate, and `file1.html` to `fileN.html` are the HTML files you want to use for the book.

The `--book` option tells *HTMLDOC* that you want to generate a book from the HTML file(s) you specified. *HTMLDOC* will build a table of contents for the book using the heading elements (`H1`, `H2`, etc.) in your HTML files. It will also add a title page using the document `TITLE` text and other `META` information you supply in your HTML files.

# Setting the Title File

The `--titlefile` option sets the HTML file or image to use on the title page:

```
% htmldoc --titlefile filename.bmp ... ENTER
% htmldoc --titlefile filename.gif ... ENTER
% htmldoc --titlefile filename.jpg ... ENTER
% htmldoc --titlefile filename.png ... ENTER
% htmldoc --titlefile filename.html ... ENTER
```

*HTMLDOC* supports BMP, GIF, JPEG, and PNG images, as well as generic HTML text you supply for the title page(s).

# Chapter 5 – Using HTMLDOC on a Web Server

This chapter describes how to interface HTMLDOC to your web server using CGI scripts and programs.

## The Basics

*HTMLDOC* can be used in a variety of ways to generate formatted reports on a web server. The most common way is to combine *HTMLDOC* with a CGI script or program and send the output to the HTTP client.

To make this work the CGI script or program must send the appropriate HTTP attributes, the required empty line to signify the beginning of the document, and then execute the *HTMLDOC* program to generate the HTML, PostScript, or PDF file as needed.

Another way to generate PDF files from your reports is to use *HTMLDOC* as a "portal" application. When used as a portal, *HTMLDOC* automatically retrieves the named document or report from your server and passes a PDF version to the web browser. See the next sections for more information.

> **WARNING:**
>
> Passing information directly from the web browser to HTMLDOC can potentially expose your system to security risks. Always be sure to "sanitize" any input from the web browser so that filenames, URLs, and options passed to HTMLDOC are not acted on by the shell program.

# Calling HTMLDOC from a Shell Script

Shell scripts are probably the easiest to work with, but are normally limited to GET type requests. Here is a script called *topdf* that acts as a portal, converting the named file to PDF:

```
#!/bin/sh
#
# Sample "portal" script to convert the named HTML file to PDF on-the-fly.
#
# Usage: http://www.domain.com/path/topdf/path/filename.html
#

#
# The "options" variable contains any options you want to pass to HTMLDOC.
#

options="-t pdf --webpage --header ... --footer ..."

#
# Tell the browser to expect a PDF file...
#

echo "Content-Type: application/pdf"
echo ""

#
# Run HTMLDOC to generate the PDF file...
#

htmldoc $options http://${SERVER_NAME}:${SERVER_PORT}$PATH_INFO
```

Users of this CGI would reference the URL "http://www.domain.com/topdf.cgi/index.html" to generate a PDF file of the site's home page.

The *options* variable in the script can be set to use any supported command-line option for HTMLDOC; for a complete list see Chapter 8 − Command-Line Reference.

## Calling HTMLDOC from Perl

Perl scripts offer the ability to generate more complex reports, pull data from databases, etc. The easiest way to interface Perl scripts with *HTMLDOC* is to write a report to a temporary file and then execute *HTMLDOC* to generate the PDF file.

Here is a simple Perl subroutine that can be used to write a PDF report to the HTTP client:

```
sub topdf(filename);

sub topdf {
    # Get the filename argument...
    my $filename = shift;

    # Make stdout unbuffered...
    select(STDOUT); $| = 1;

    # Write the content type to the client...
    print "Content-Type: application/pdf\n\n";

    # Run HTMLDOC to provide the PDF file to the user...
    system "htmldoc -t pdf --quiet --webpage $filename";
}
```

## Calling HTMLDOC from PHP

PHP is quickly becoming the most popular server-side scripting language available. PHP provides a `passthru()` function that can be used to run *HTMLDOC*. This combined with the `header()` function can be used to provide on-the-fly reports in PDF format.

Here is a simple PHP function that can be used to convert a HTML report to PDF and send it to the HTTP client:

```
function topdf($filename, $options = "") {
    # Write the content type to the client...
    header("Content-Type: application/pdf");
    flush();

    # Run HTMLDOC to provide the PDF file to the user...
    passthru("htmldoc -t pdf --quiet --jpeg --webpage $options \'$filename\'");
}
```

The function accepts a filename and an optional "options" string for specifying the header, footer, fonts, etc.

To prevent malicious users from passing in unauthorized characters into this function, the following function can be used to verify that the URL/filename does not contain any characters that might be interpreted by the shell:

```
function bad_url($url) {
    // See if the URL starts with http: or https:...
    if (strncmp($url, "http://", 7) != 0 &&
        strncmp($url, "https://", 8) != 0) {
        return 1;
    }
```

```
    // Check for bad characters in the URL...
    $len = strlen($url);
    for ($i = 0; $i < $len; $i ++) {
        if (!strchr("~_*()/:%?+-&@;=,$.", $url[$i]) &&
            !ctype_alnum($url[$i])) {
            return 1;
        }
    }

    return 0;
}
```

Another method is to use the `escapeshellarg()` function provided with PHP 4.0.3 and higher to generate a quoted shell argument for HTMLDOC.

To make a "portal" script, add the following code to complete the example:

```
global $SERVER_NAME;
global $SERVER_PORT;
global $PATH_INFO;
global $QUERY_STRING;

if ($QUERY_STRING != "") {
    $url = "http://${SERVER_NAME}:${SERVER_PORT}${PATH_INFO}?${QUERY_STRING}";
} else {
    $url = "http://${SERVER_NAME}:${SERVER_PORT}$PATH_INFO";
}

if (bad_url($url)) {
  print("<HTML><HEAD><TITLE>Bad URL</TITLE></HEAD>\n"
       ."<BODY><H1>Bad URL</H1>\n",
       ."<P>The URL <B><TT>$url</TT></B> is bad.</P>\n"
       ."</BODY></HTML>\n");
} else {
  topdf($url);
}
```

## Calling HTMLDOC from C

C programs offer the best flexibility and easily support on−the−fly report generation without the need for temporary files.

Here are some simple C functions that can be used to generate a PDF report to the HTTP client from a temporary file or pipe:

```
#include <stdio.h>
#include <stdlib.h>


/* topdf() − convert a HTML file to PDF */
FILE *topdf(const char *filename)       /* HTML file to convert */
{
  char  command[1024];                  /* Command to execute */


  puts("Content−Type: application/pdf\n");

  sprintf(command, "htmldoc −t pdf −−webpage %s", filename);

  return (popen(command, "w"));
}


/* topdf2() − pipe HTML output to HTMLDOC for conversion to PDF */
FILE *topdf2(void)
{
  puts("Content−Type: application/pdf\n");
  return (popen("htmldoc −t pdf −−webpage −", "w"));
}
```

# Calling HTMLDOC from Java

Java programs are a portable way to add PDF support to your web server. Here is a class called *htmldoc* that acts as a portal, converting the named file to PDF. It can also be called by your Java servlets to process an HTML file and send the result to the client in PDF format:

```
class htmldoc
{
  // Convert named file to PDF on stdout...
  public static int topdf(String filename)// I − Name of file to convert
  {
    String              command;        // Command string
    Process             process;        // Process for HTMLDOC
    Runtime             runtime;        // Local runtime object
    java.io.InputStream input;          // Output from HTMLDOC
    byte                buffer [];      // Buffer for output data
    int                 bytes;          // Number of bytes


    // First tell the client that we will be sending PDF...
    System.out.print("Content−type: application/pdf\n\n");

    // Construct the command string
    command = "htmldoc −−quiet −−jpeg −−webpage −t pdf −−left 36 " +
              "−−header .t. −−footer .1. " + filename;

    // Run the process and wait for it to complete...
    runtime = Runtime.getRuntime();

    try
    {
      // Create a new HTMLDOC process...
      process = runtime.exec(command);

      // Get stdout from the process and a buffer for the data...
      input  = process.getInputStream();
      buffer = new byte[8192];

      // Read output from HTMLDOC until we have it all...
      while ((bytes = input.read(buffer)) > 0)
        System.out.write(buffer, 0, bytes);

      // Return the exit status from HTMLDOC...
      return (process.waitFor());
    }
    catch (Exception e)
    {
      // An error occurred − send it to stderr for the web server...
      System.err.print(e.toString() + " caught while running:\n\n");
      System.err.print("    " + command + "\n");
      return (1);
    }
  }

  // Main entry for htmldoc class
  public static void main(String[] args)// I − Command−line args
  {
    String      server_name,            // SERVER_NAME env var
                server_port,            // SERVER_PORT env var
```

```
                path_info,              // PATH_INFO env var
                query_string,           // QUERY_STRING env var
                filename;               // File to convert


    if ((server_name = System.getProperty("SERVER_NAME")) != null &&
        (server_port = System.getProperty("SERVER_PORT")) != null &&
        (path_info = System.getProperty("PATH_INFO")) != null)
    {
      // Construct a URL for the resource specified...
      filename = "http://" + server_name + ":" + server_port + path_info;

      if ((query_string = System.getProperty("QUERY_STRING")) != null)
      {
        filename = filename + "?" + query_string;
      }
    }
    else if (args.length == 1)
    {
      // Pull the filename from the command-line...
      filename = args[0];
    }
    else
    {
      // Error - no args or env variables!
      System.err.print("Usage: htmldoc.class filename\n");
      return;
    }

    // Convert the file to PDF and send to the web client...
    topdf(filename);
  }
}
```

# Chapter 6 − HTML Reference

This chapter defines all of the HTML elements and attributes that are recognized and supported by *HTMLDOC*.

## General Usage

There are two types of HTML files − structured documents using headings (H1, H2, etc.) which *HTMLDOC* calls "books", and unstructured documents that do not use headings which *HTMLDOC* calls "web pages".

A very common mistake is to try converting a web page using:

```
htmldoc -f filename.pdf filename.html
```

which will likely produce a PDF file with no pages. To convert web page files you **must** use the `--webpage` option at the command−line or choose *Web Page* in the input tab of the GUI.

***HTMLDOC* does not support HTML 4.0 elements, attributes, stylesheets, or scripting.**

# Elements

The following HTML elements are recognized by *HTMLDOC*:

| Element | Version | Supported? | Notes |
|---|---|---|---|
| !DOCTYPE | 3.0 | Yes | DTD is ignored |
| A | 1.0 | Yes | See Below |
| ACRONYM | 2.0 | Yes | No font change |
| ADDRESS | 2.0 | Yes | |
| AREA | 2.0 | No | |
| B | 1.0 | Yes | |
| BASE | 2.0 | No | |
| BASEFONT | 1.0 | No | |
| BIG | 2.0 | Yes | |
| BLINK | 2.0 | No | |
| BLOCKQUOTE | 2.0 | Yes | |
| BODY | 1.0 | Yes | |
| BR | 2.0 | Yes | |
| CAPTION | 2.0 | Yes | See Below |
| CENTER | 2.0 | Yes | |
| CITE | 2.0 | Yes | Italic/Oblique |
| CODE | 2.0 | Yes | Courier |
| DD | 2.0 | Yes | |
| DEL | 2.0 | Yes | Strikethrough |
| DFN | 2.0 | Yes | Helvetica |
| DIR | 2.0 | Yes | |
| DIV | 3.2 | Yes | |
| DL | 2.0 | Yes | |
| DT | 2.0 | Yes | Italic/Oblique |
| EM | 2.0 | Yes | Italic/Oblique |
| EMBED | 2.0 | Yes | HTML Only |
| FONT | 2.0 | Yes | See Below |
| FORM | 2.0 | No | |
| FRAME | 3.2 | No | |
| FRAMESET | 3.2 | No | |
| H1 | 1.0 | Yes | Boldface, See Below |
| H2 | 1.0 | Yes | Boldface, See Below |
| H3 | 1.0 | Yes | Boldface, See Below |
| H4 | 1.0 | Yes | Boldface, See Below |

| | | | |
|---|---|---|---|
| H5 | 1.0 | Yes | Boldface, See Below |
| H6 | 1.0 | Yes | Boldface, See Below |
| HEAD | 1.0 | Yes | |
| HR | 1.0 | Yes | See Below |
| HTML | 1.0 | Yes | |
| I | 1.0 | Yes | |
| IMG | 1.0 | Yes | See Below |
| INPUT | 2.0 | No | |
| INS | 2.0 | Yes | Underline |
| ISINDEX | 2.0 | No | |
| KBD | 2.0 | Yes | Courier Bold |
| LI | 2.0 | Yes | |
| LINK | 2.0 | No | |
| MAP | 2.0 | No | |
| MENU | 2.0 | Yes | |
| META | 2.0 | Yes | See Below |
| MULTICOL | N3.0 | No | |
| NOBR | 1.0 | No | |
| NOFRAMES | 3.2 | No | |
| OL | 2.0 | Yes | |
| OPTION | 2.0 | No | |
| P | 1.0 | Yes | |
| PRE | 1.0 | Yes | |
| S | 2.0 | Yes | Strikethrough |
| SAMP | 2.0 | Yes | Courier |
| SCRIPT | 2.0 | No | |
| SELECT | 2.0 | No | |
| SMALL | 2.0 | Yes | |
| SPACER | N3.0 | Yes | |
| STRIKE | 2.0 | Yes | |
| STRONG | 2.0 | Yes | Boldface Italic/Oblique |
| SUB | 2.0 | Yes | Reduced Fontsize |
| SUP | 2.0 | Yes | Reduced Fontsize |
| TABLE | 2.0 | Yes | See Below |
| TD | 2.0 | Yes | |
| TEXTAREA | 2.0 | No | |
| TH | 2.0 | Yes | Boldface Center |
| TITLE | 2.0 | Yes | |

| | | | |
|---|---|---|---|
| TR | 2.0 | Yes | |
| TT | 2.0 | Yes | Courier |
| U | 1.0 | Yes | |
| UL | 2.0 | Yes | |
| VAR | 2.0 | Yes | Helvetica Oblique |
| WBR | 1.0 | No | |

## Comments

*HTMLDOC* supports many special HTML comments to initiate page breaks, set the header and footer text, and control the current media options:

`<!-- FOOTER LEFT "foo" -->`
: Sets the left footer text; the test is applied to the current page if empty, or the next page otherwise.

`<!-- FOOTER CENTER "foo" -->`
: Sets the center footer text; the test is applied to the current page if empty, or the next page otherwise.

`<!-- FOOTER RIGHT "foo" -->`
: Sets the right footer text; the test is applied to the current page if empty, or the next page otherwise.

`<!-- HALF PAGE -->`
: Break to the next half page.

`<!-- HEADER LEFT "foo" -->`
: Sets the left header text; the test is applied to the current page if empty, or the next page otherwise.

`<!-- HEADER CENTER "foo" -->`
: Sets the center header text; the test is applied to the current page if empty, or the next page otherwise.

`<!-- HEADER RIGHT "foo" -->`
: Sets the right header text; the test is applied to the current page if empty, or the next page otherwise.

`<!-- MEDIA BOTTOM nnn -->`
: Sets the bottom margin of the page. The "nnn" string can be any standard measurement value, e.g. 0.5in, 36, 12mm, etc. Breaks to a new page if the current page is already marked.

`<!-- MEDIA COLOR "foo" -->`
: Sets the media color attribute for the page. The "foo" string is any color name that is supported by the printer, e.g. "Blue", "White", etc. Breaks to a new page or sheet if the current page is already marked.

`<!-- MEDIA DUPLEX NO -->`
: Chooses single-sided printing for the page; breaks to a new page or sheet if the current page is already marked.

`<!-- MEDIA DUPLEX YES -->`
: Chooses double-sided printing for the page; breaks to a new sheet if the current page is already marked.

`<!-- MEDIA LANDSCAPE NO -->`
: Chooses portrait orientation for the page; breaks to a new page if the current page is already marked.

`<!-- MEDIA LANDSCAPE YES -->`
: Chooses landscape orientation for the page; breaks to a new page if the current page is already marked.

`<!-- MEDIA LEFT nnn -->`
: Sets the left margin of the page. The "nnn" string can be any standard measurement value, e.g. 0.5in, 36, 12mm, etc. Breaks to a new page if the current page is already marked.

`<!-- MEDIA POSITION nnn -->`
: Sets the media position attribute (input tray) for the page. The "nnn" string is an integer that usually

specifies the tray number. Breaks to a new page or sheet if the current page is already marked.

*<!-- MEDIA RIGHT nnn -->*

Sets the right margin of the page. The "nnn" string can be any standard measurement value, e.g. 0.5in, 36, 12mm, etc. Breaks to a new page if the current page is already marked.

*<!-- MEDIA SIZE foo -->*

Sets the media size to the specified size. The "foo" string can be "Letter", "Legal", "Universal", or "A4" for standard sizes or "WIDTHxHEIGHTunits" for custom sizes, e.g. "8.5x11in"; breaks to a new page or sheet if the current page is already marked.

*<!-- MEDIA TOP nnn -->*

Sets the top margin of the page. The "nnn" string can be any standard measurement value, e.g. 0.5in, 36, 12mm, etc. Breaks to a new page if the current page is already marked.

*<!-- MEDIA TYPE "foo" -->*

Sets the media type attribute for the page. The "foo" string is any type name that is supported by the printer, e.g. "Plain", "Glossy", etc. Breaks to a new page or sheet if the current page is already marked.

*<!-- NEED length -->*

Break if there is less than `length` units left on the current page. The `length` value defaults to lines of text but can be suffixed by `in`, `mm`, or `cm` to convert from the corresponding units.

*<!-- NEW PAGE -->*

Break to the next page.

*<!-- NEW SHEET -->*

Break to the next sheet.

*<!-- PAGE BREAK -->*

Break to the next page.

## Header/Footer Strings

The `HEADER` and `FOOTER` comments allow you to set an arbitrary string of text for the left, center, and right headers and footers. Each string consists of plain text; special values or strings can be inserted using the dollar sign ($):

*$$*

Inserts a single dollar sign in the header.

*CHAPTER*

Inserts the current chapter heading.

*$CHAPTERPAGE*

*$CHAPTERPAGE(format)*

Inserts the current page number within a chapter or file. When a format is specified, uses that numeric format (1 = decimal, i = lowercase roman numerals, I = uppercase roman numerals, a = lowercase ascii, A = uppercase ascii) for the page numbers.

*$CHAPTERPAGES*

*$CHAPTERPAGES(format)*

Inserts the total page count within a chapter or file. When a format is specified, uses that numeric format (1 = decimal, i = lowercase roman numerals, I = uppercase roman numerals, a = lowercase ascii, A = uppercase ascii) for the page count.

*$DATE*

Inserts the current date.

*$HEADING*

Inserts the current heading.

*$PAGE*

*$PAGE(format)*

Inserts the current page number. When a format is specified, uses that numeric format (1 = decimal, i = lowercase roman numerals, I = uppercase roman numerals, a = lowercase ascii, A = uppercase ascii) for the page numbers.

*$PAGES*

*$PAGES(format)*

Inserts the total page count. When a format is specified, uses that numeric format (1 = decimal, i = lowercase roman numerals, I = uppercase roman numerals, a = lowercase ascii, A = uppercase ascii) for the page count.

*$TIME*

Inserts the current time.

*$TITLE*

Inserts the document title.

## FONT Attributes

Limited typeface specification is currently supported to ensure portability across platforms and for older PostScript printers:

| Requested Font | Actual Font |
|---|---|
| Arial | Helvetica |
| Courier | Courier |
| Helvetica | Helvetica |
| Monospace | Courier |
| Sans−Serif | Helvetica |
| Serif | Times |
| Symbol | Symbol |
| Times | Times |

All other unrecognized typefaces are silently ignored.

## Headings

Currently *HTMLDOC* supports a maximum of 10000 headings and 100 chapters. These limits can be increased by changing the constants in the *config.h* file included with the source code.

All chapters start with a top−level heading (H1) markup. Any headings within a chapter must be of a lower level (H2 to H6). Each chapter starts a new page or the next odd−numbered page if duplexing is selected.

The headings you use within a chapter must start at level 2 (H2). If you skip levels the heading will be shown under the last level that was known. For example, if you use the following hierarchy of headings:

```
<H1>Chapter Heading</H1>
...
<H2>Section Heading 1</H2>
...
<H2>Section Heading 2</H2>
...
<H3>Sub−Section Heading 1</H3>
```

```
...
<H4>Sub−Sub−Section Heading 1</H4>
...
<H4>Sub−Sub−Section Heading 2</H4>
...
<H3>Sub−Section Heading 2</H3>
...
<H2>Section Heading 3</H2>
...
<H4>Sub−Sub−Section Heading 3</H4>
...
```

the table−of−contents that is generated will show:

**Chapter Heading**

- ♦ Section Heading 1
- ♦ Section Heading 2
  - ◊ Sub−Section Heading 1
    - · Sub−Sub−Section Heading 1
    - · Sub−Sub−Section Heading 2
  - ◊ Sub−Section Heading 2
    - · Sub−Sub−Section Heading 3
- ♦ Section Heading 3

### Numbered Headings

When the numbered headings option is enabled, *HTMLDOC* recognizes the following additional attributes for all heading elements:

*VALUE="#"*
: Specifies the starting value for this heading level (default is "1" for all new levels).

*TYPE="1"*
: Specifies that decimal numbers should be generated for this heading level.

*TYPE="a"*
: Specifies that lowercase letters should be generated for this heading level.

*TYPE="A"*
: Specifies that uppercase letters should be generated for this heading level.

*TYPE="i"*
: Specifies that lowercase roman numerals should be generated for this heading level.

*TYPE="I"*
: Specifies that uppercase roman numerals should be generated for this heading level.

## Images

*HTMLDOC* supports loading of BMP, GIF, JPEG, and PNG image files. EPS and other types of image files are not supported at this time.

## Links

Currently *HTMLDOC* supports a maximum of 20000 links within a document. This limit can be increased by changing the constant in the *config.h* file included with the source code.

External URL and internal (`#target` and `filename.html`) links are fully supported for HTML and PDF output.

When generating PDF files, local PDF file links will be converted to external file links for the PDF viewer instead of URL links. That is, you can directly link to another local PDF file from your HTML document with:

```
<A HREF="filename.pdf">...</A>
```

## META Attributes

*HTMLDOC* supports the following `META` attributes for the title page and document information:

*<META NAME="AUTHOR" CONTENT="...">*
: Specifies the document author.

*<META NAME="COPYRIGHT" CONTENT="...">*
: Specifies the document copyright.

*<META NAME="DOCNUMBER" CONTENT="...">*
: Specifies the document number.

*<META NAME="GENERATOR" CONTENT="...">*
: Specifies the application that generated the HTML file.

*<META NAME="KEYWORDS" CONTENT="...">*
: Specifies document search keywords.

## Page Breaks

*HTMLDOC* supports four new page comments to specify page breaks. In addition, the older `BREAK` attribute is still supported by the `HR` element:

```
<HR BREAK>
```

Support for the `BREAK` attribute is deprecated and will be removed in a future release of *HTMLDOC*.

## Tables

Currently *HTMLDOC* supports a maximum of 200 columns within a single table. This limit can be increased by changing the `MAX_COLUMNS` constant in the *config.h* file included with the source code. *HTMLDOC* supports HTML 3.0 tables with the following exceptions:

- The `CAPTION` element is always shown at the top of the table.

***HTMLDOC* does not support HTML 4.0 table elements or attributes, such as `TBODY`, `THEAD`, `TFOOT`, or `RULES`.**

# Chapter 7 – GUI Reference

This chapter describes all of the GUI controls in *HTMLDOC*.

## The HTMLDOC GUI

The *HTMLDOC* GUI (Figures 7−1 through 7−10) is contained in a single window showing the input, output, and generation options. At the bottom are buttons to load, save, and generate documents.

### Document File Operations

*HTMLDOC* stores the HTML files, settings, and options in `.BOOK` files. The buttons on the bottom of the *HTMLDOC* window allow you to manage these files and generate formatted documents.

#### New

The *New* button starts a new document. A confirmation dialog will appear if you have not saved the changes to the existing document.

#### Open...

The *Open...* button retrieves a document that you have saved previously. A file chooser dialog is displayed that allows you to pick an existing book file.

## Save

The *Save* button saves the current document. A file chooser dialog is displayed if there is no filename assigned to the current document.

**Note:** Saving a document is not the same as *generating* a document. The book files saved to disk by the *Save* and *Save As...* buttons are ***not*** the final HTML, PDF, or PostScript output files. You generate those files by clicking on the *Generate* button.

## Save As...

The *Save As...* button saves the current document to a new file. A file chooser dialog is displayed to allow you to specify the new document filename.

**Note:** Saving a document is not the same as *generating* a document. The book files saved to disk by the *Save* and *Save As...* buttons are ***not*** the final HTML, PDF, or PostScript output files. You generate those files by clicking on the *Generate* button.

## Generate

The *Generate* button generates the current document, creating the specified HTML, PDF, or PostScript file(s) as needed. The progress meter at the bottom of the window will show the progress as each page or file is formatted and written.

**Note:** Generating a document is not the same as ***saving*** a document. To save the current HTML files and settings in the *HTMLDOC* GUI, click on the *Save* or *Save As...* buttons instead.

## Close

The *Close* button closes the *HTMLDOC* window.

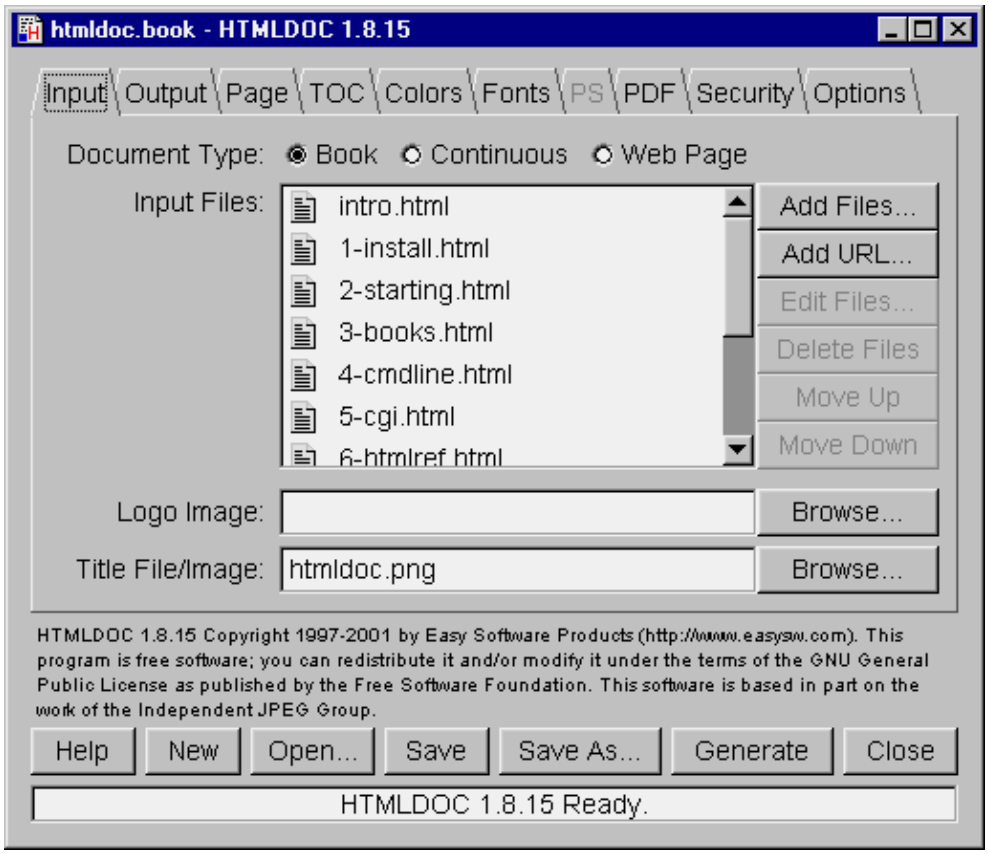



Figure 7−1 − The Input Tab

# The Input Tab

The input tab (Figure 7−1) lists all of the HTML source files that are used to generate the document. You also specify the type of document (book or web page) and the title and logo images in this tab.

## Document Type

The *Book* radio button specifies that the input files are structured with headings. The *Continuous* radio button specifies unstructured files without page breaks between each file. The *Web Page* radio button specifies unstructured files with page breaks between each file.

## Input Files

The *Input Files* list shows all of the HTML input files that will be used to produce the document. Double−click on files to edit them.

## Add Files...

The *Add Files...* button displays the file chooser dialog, allowing you to select one or more HTML files to include in the document.

## Edit Files...

The *Edit Files...* button starts the specified editor program to edit the files selected in the *Input Files* list. Select one or more files in the *Input Files* list to enable the *Edit Files...* button.

## Delete Files

The *Delete Files* button removes the selected files from the *Input Files* list. Select one or more files in the *Input Files* list to enable the *Delete Files* button.

The *Delete Files* button only removes the files from the *Input Files* list. The files are ***not*** removed from disk.

## Move Up

The *Move Up* button moves the selected files in the *Input Files* list up one line in the list. To enable the *Move Up* button select one or more files in the *Input Files* list.

## Move Down

The *Move Down* button moves the selected files in the *Input Files* list down one line in the list. To enable the *Move Down* button select one or more files in the *Input Files* list.

## Logo Image

The *Logo Image* field contains the filename for an image to be shown in the header or footer of pages, and in the navigation bar of HTML files.

Click on the *Browse...* button to select a logo image file using the file chooser dialog.

## Title File/Image

The *Title File/Image* field contains the filename for an image to be shown on the title page, or for a HTML file to be used for the title page(s).

Click on the *Browse...* button to select a title file using the file chooser dialog.

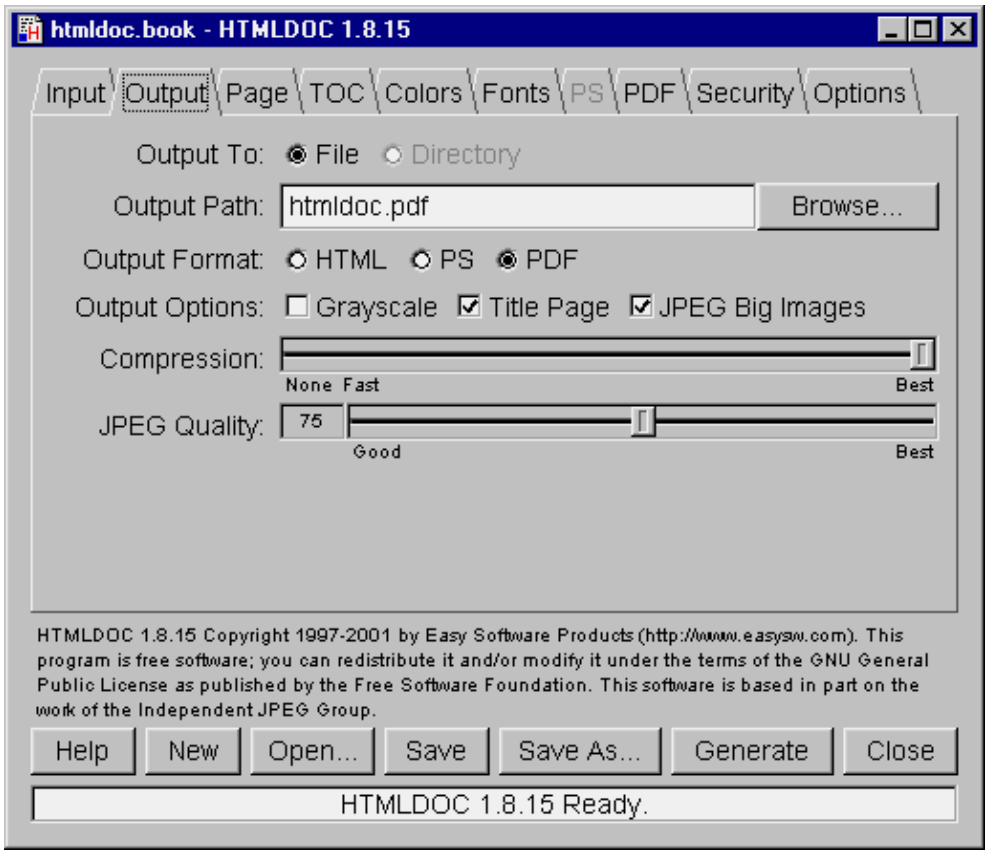



Figure 7−2 − The Output Tab

# The Output Tab

The output tab (Figure 7−2) specifies where your document will be generated, the output format, and some of the generic output options.

## Output To

The *File* radio button selects output to a single file. The *Directory* radio button selects output to multiple files in the named directory.

*Directory* output is not available when generating PDF files.

## Output Path

The *Output Path* field contains the output directory or filename. Click on the *Browse...* button to choose an output file using the file chooser dialog.

## Output Format

The *HTML* radio button selects HTML output, the *PS* radio button selects PostScript output, and the *PDF* radio button selects PDF output.

### Output Options

The *Grayscale* check box selects grayscale output for PostScript and PDF files. The *Title Page* check box specifies that a title page should be generated for the document. The *JPEG Big Images* check box specifies that JPEG compression should be applied to continuous−tone images.

### Compression

The *Compression* slider controls the amount of compression that is used when writing PDF or Level 3 PostScript output.

**Note:** *HTMLDOC* uses Flate compression, which is not encumbered by patents and is also used by the popular PKZIP and gzip programs. Flate is a lossless compression algorithm (that is, you get back exactly what you put in) that performs very well on indexed images and text.

### JPEG Quality

The *JPEG Quality* slider controls the quality level used when writing continuous−tone images with JPEG compression.

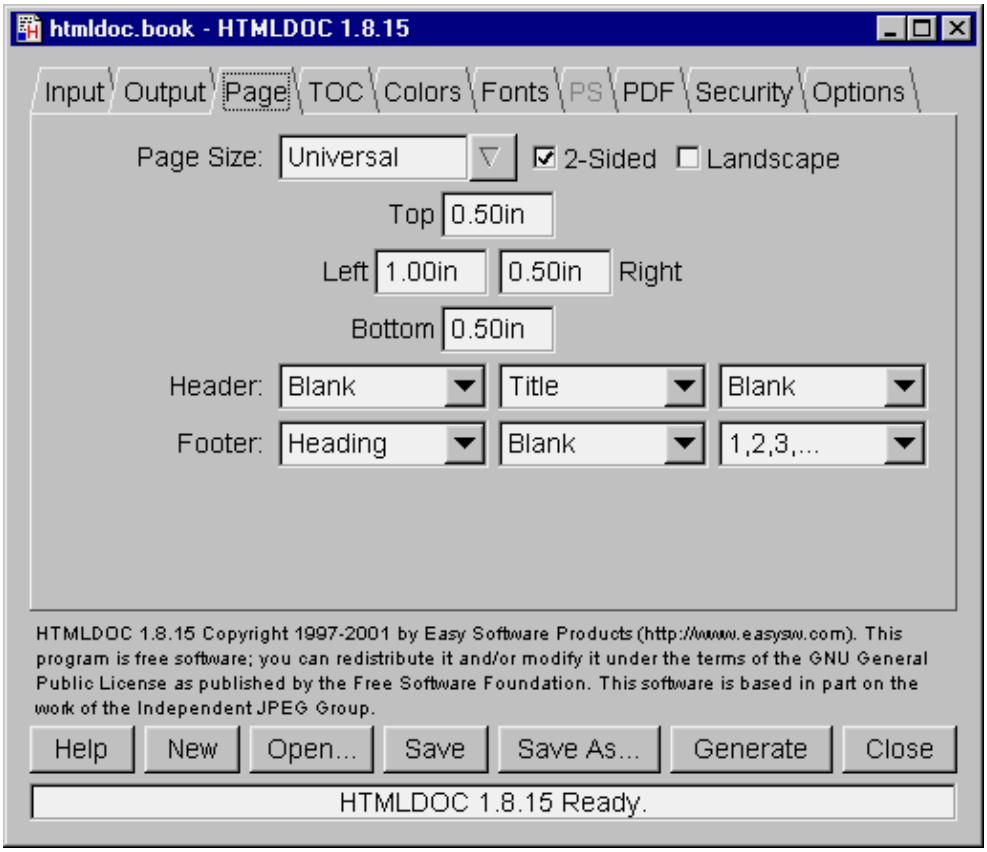



Figure 7−3 − The Page Tab

## The Page Tab

The page tab (Figure 7−3) defines the page header, footer, size, and margins for PostScript and PDF output.

## Page Size

The *Page Size* field contains the current page size. Click on the arrow button to choose a standard page size.

*HTMLDOC* supports the following standard page size names:

- Letter − 8.5x11in (216x279mm)
- A4 − 8.27x11.69in (210x297mm)
- Universal − 8.27x11in (210x279mm)

Click in the *Page Size* field and enter the page width and length separated by the letter "x" to select a custom page size. Append the letters "in" for inches, "mm" for millimeters, or "cm" for centimeters.

## 2−Sided

Click in the *2−Sided* check box to select 2−sided (duplexed) output.

## Landscape

Click in the *Landscape* check box to select landscape output.

## Top, Left, Right, and Bottom

Click in the *Top*, *Left*, *Right*, and *Bottom* fields and enter the new margin values to change them. Append the letters "in" for inches, "mm" for millimeters, or "cm" for centimeters.

## Header and Footer

Select the desired text in each of the option buttons to customize the header and footer for the document/body pages. The left−most option buttons set the text that is left−justified, while the middle buttons set the text that is centered and the right buttons set the text that is right−justified. Each choice corresponds to the following text:

| Choice | Description |
|---|---|
| Blank | The field should be blank. |
| Title | The field should contain the document title. |
| Chapter Title | The field should contain the current chapter title. |
| Heading | The field should contain the current heading. |
| Logo | The field should contain the logo image. |
| 1,2,3,... | The field should contain the current page number in decimal format (1, 2, 3, ...) |
| i,ii,iii,... | The field should contain the current page number in lowercase roman numerals (i, ii, iii, ...) |
| I,II,III,... | The field should contain the current page number in uppercase roman numerals (I, II, III, ...) |
| a,b,c,... | The field should contain the current page number using lowercase letters. |

| | |
|---|---|
| A,B,C,... | The field should contain the current page number using UPPERCASE letters. |
| Chapter Page | The field should contain the current chapter page number. |
| 1/N,2/N,... | The field should contain the current and total number of pages (n/N). |
| 1/C,2/C,... | The field should contain the current and total number of pages in the chapter (n/N). |
| Date | The field should contain the current date (formatted for the current locale). |
| Time | The field should contain the current time (formatted for the current locale). |
| Date + Time | The field should contain the current date and time (formatted for the current locale). |

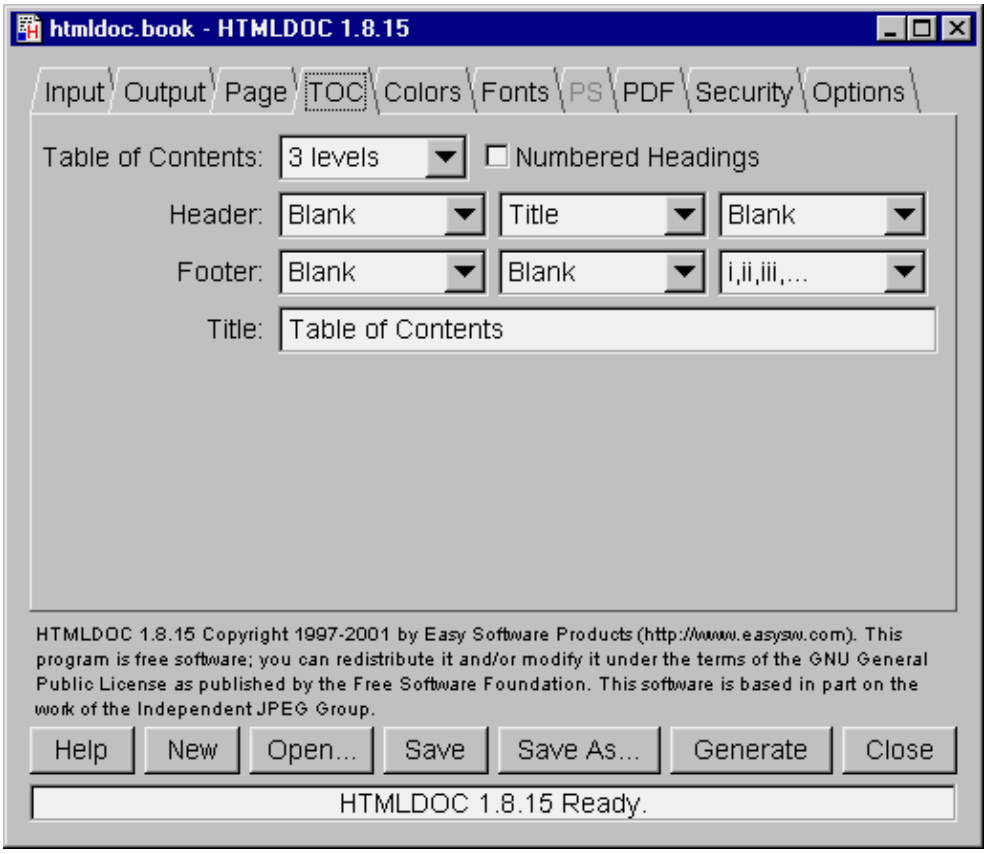



Figure 7−4 − The TOC Tab

# The TOC Tab

The TOC tab (Figure 7−4) defines the table−of−contents options.

## Table of Contents

Select the desired number of levels from the *Table of Contents* option button.

### Numbered Headings

Click in the *Numbered Headings* check box to automatically number the headings in the document.

### Header and Footer

Select the desired text in each of the option buttons to customize the header and footer for the tables−of−contents pages. The left−most option buttons set the text that is left−justified, while the middle buttons set the text that is centered and the right buttons set the text that is right−justified.

### Title

Enter the desired title for the table−of−contents in the *Title* field.

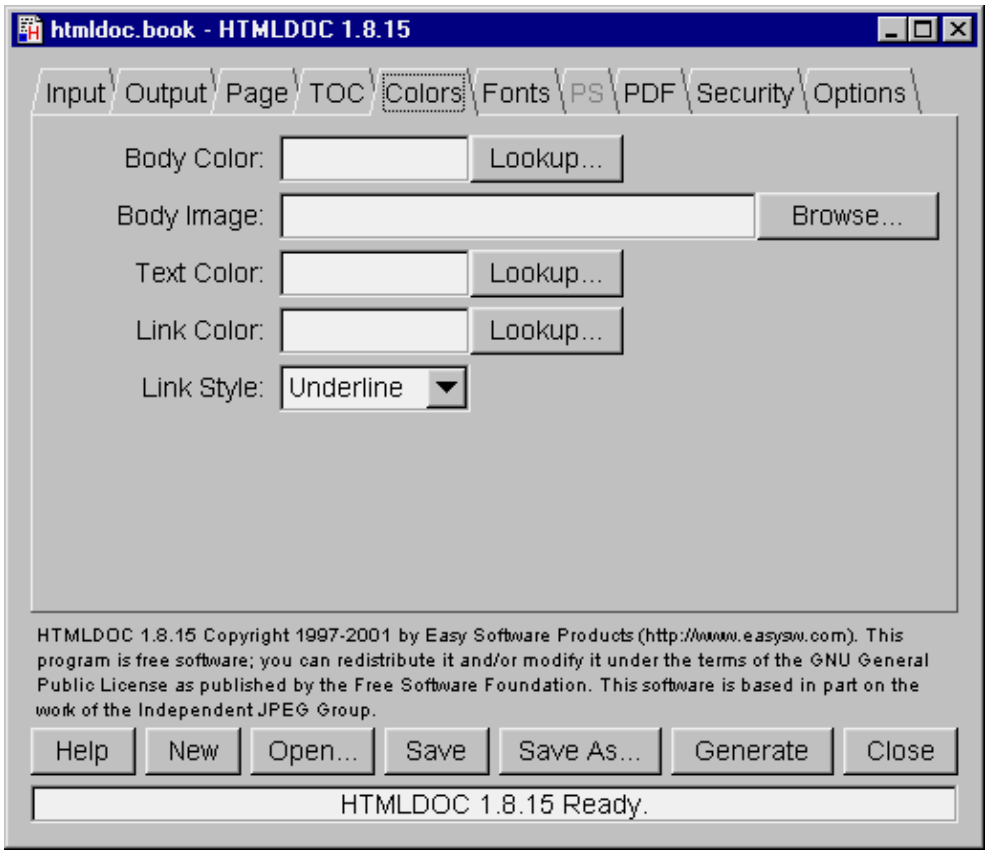



Figure 7−5 − The Colors Tab

## The Colors Tab

The colors tab (Figure 7−5) defines the color and image information that is used for the entire document.

### Body Color

The *Body Color* field specifies the default background color. It can be a standard HTML color name or a hexadecimal RGB color of the form #RRGGBB. Click on the *Lookup...* button to pick the color graphically.

### Body Image

The *Body Image* field specifies the default background image. Click on the *Browse...* button to pick the background image using the file chooser.

### Text Color

The *Text Color* field specifies the default text color. It can be a standard HTML color name or a hexadecimal RGB color of the form #RRGGBB. Click on the *Lookup...* button to pick the color graphically.

### Link Color

The *Link Color* field specifies the default link color. It can be a standard HTML color name or a hexadecimal RGB color of the form #RRGGBB. Click on the *Lookup...* button to pick the color graphically.

### Link Style

The *Link Style* chooser specifies the default link decoration.

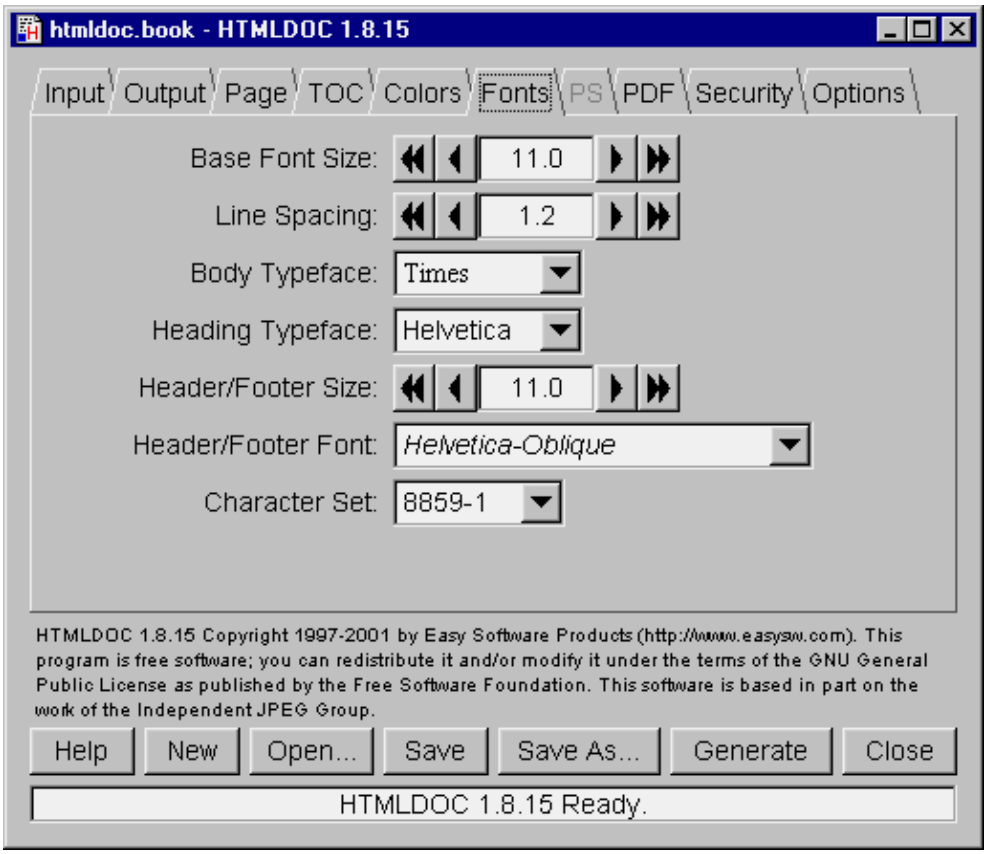



Figure 7−6 − The Fonts Tab

## The Fonts Tab

The fonts tab (Figure 7−6) defines the fonts and character set used by the document.

## Base Font Size

The *Base Font Size* field specifies the size of normal text in the document in points (1 point = 1/72nd inch). Click on the single arrow buttons to decrease or increase the size by 1/10th point or on the double arrow buttons to decrease or increase the size by whole points.

## Line Spacing

The *Line Spacing* field specifies the spacing between lines as a multiple of the base font size. Click on the single arrow buttons to decrease or increase the size by 10ths or on the double arrow buttons to decrease or increase the size by whole numbers.

## Body Typeface

The *Body Typeface* option button specifies the typeface to use for normal text. Click on the option button to select a typeface.

## Heading Typeface

The *Heading Typeface* option button specifies the typeface to use for headings. Click on the option button to select a typeface.

## Header/Footer Size

The *Header/Footer Size* field specifies the size of header and footer text in the document in points (1 point = 1/72nd inch). Click on the single arrow buttons to decrease or increase the size by 1/10th point or on the double arrow buttons to decrease or increase the size by whole points.

## Header/Footer Font

The *Header/Footer Font* option button specifies the typeface and style to use for header and footer text. Click on the option button to select a typeface and style.

## Character Set

The *Character Set* option button specifies the encoding of characters in the document. Click on the option button to select a character set.

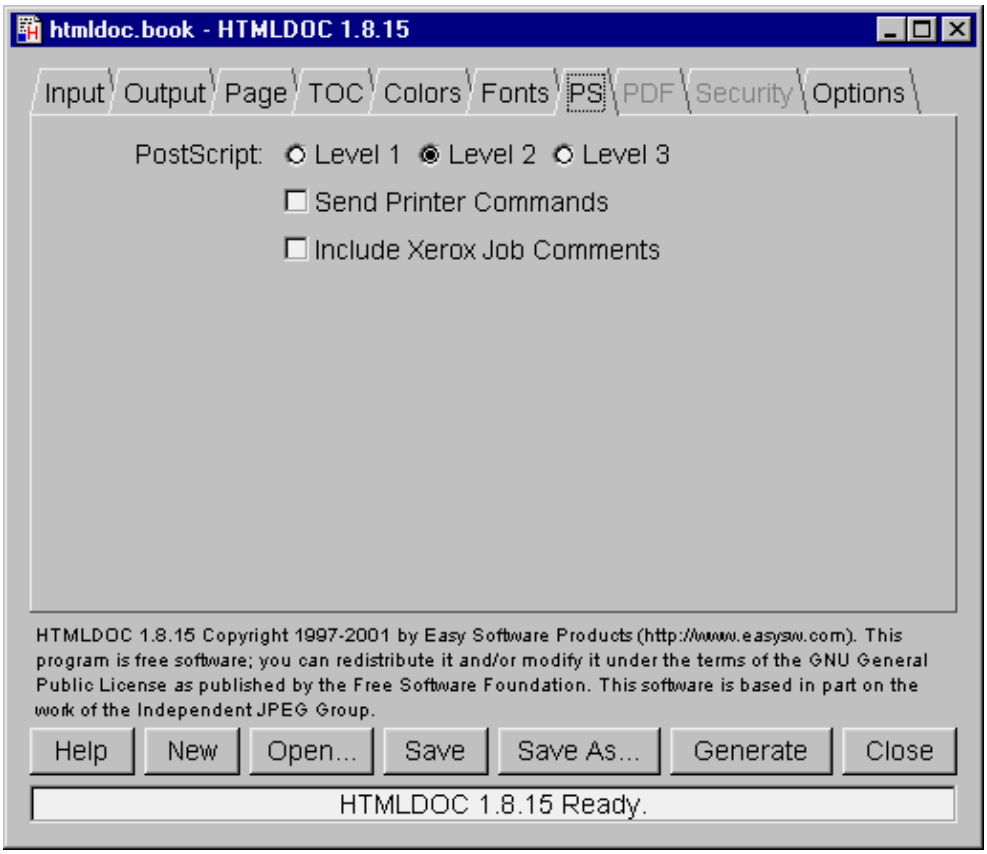



Figure 7−7 − The PS Tab

# The PS Tab

The PS tab (Figure 7−7) contains options specific to PostScript output.

## PostScript Level

Click on one of the *Level* radio buttons to select the language level to generate. PostScript Level 1 is compatible with all PostScript printers and will produce the largest output files.

PostScript Level 2 is compatible with most PostScript printers and supports printer commands and JPEG image compression.

PostScript Level 3 is compatible with only the newest PostScript printers and supports Flate image compression in addition to the Level 2 features.

## Send Printer Commands

The *Send Printer Commands* check box controls whether or not the output files contain PostScript `setpagedevice` commands for the page size and duplex settings. Click in the check box to enable or disable printer commands.

Printer commands are only available with Level 2 and 3 output and may not work with some printers.

### Include Xerox Job Comments

The *Include Xerox Job Comments* check box controls whether or not the output files contain Xerox job comments. Click in the check box to enable or disable the job comments.

Job comments are available with all levels of PostScript output.

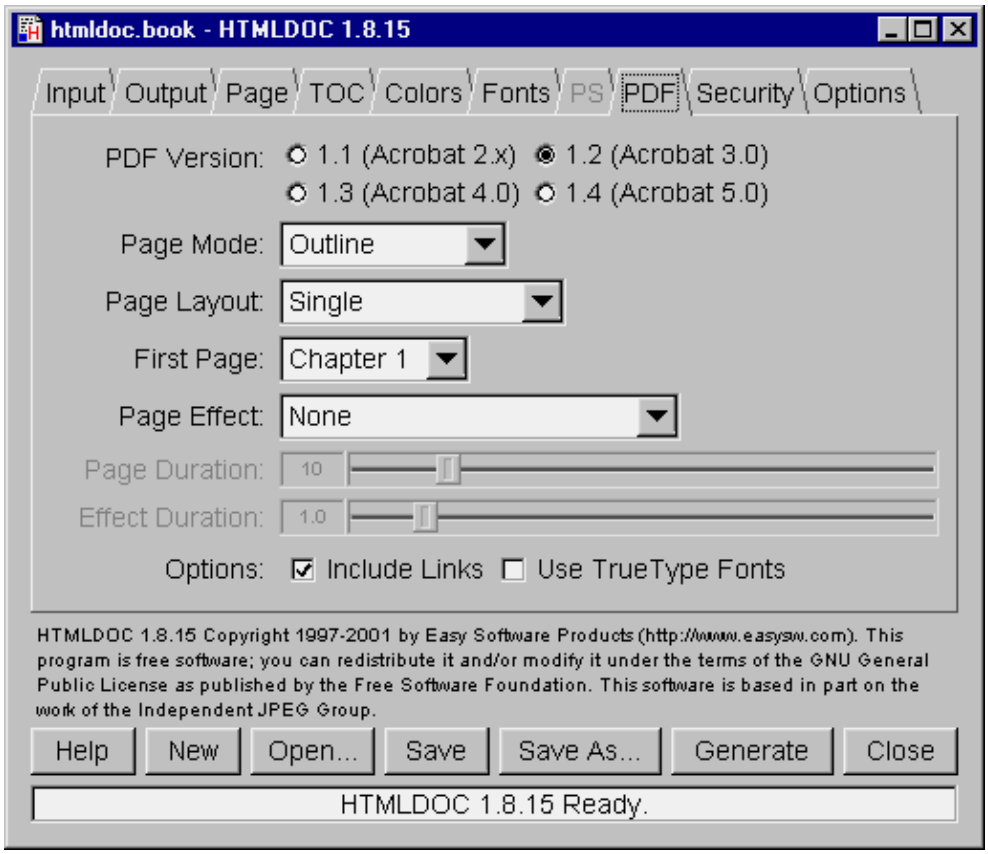



Figure 7−8 − The PDF Tab

## The PDF Tab

The PDF tab (Figure 7−8) contains settings specific to PDF output.

### PDF Version

The *PDF Version* radio buttons control what version of PDF is generated. PDF 1.3 is the most commonly supported version. Click on the corresponding radio button to set the version.

### Page Mode

The *Page Mode* option button controls the initial viewing mode for the document. Click on the option button to set the page mode.

The *Document* page mode displays only the document pages. The *Outline* page mode displays the table−of−contents outline as well as the document pages. The *Full−Screen* page mode displays the document pages on the whole screen; this mode is used primarily for presentations.

## Page Layout

The *Page Layout* option button controls the initial layout of document pages on the screen. Click on the option button to set the page layout.

The *Single* page layout displays a single page at a time. The *One Column* page layout displays a single column of pages at a time. The *Two Column Left* and *Two Column Right* page layouts display two columns of pages at a time; the first page is displayed in the left or right column as selected.

## First Page

The *First Page* option button controls the initial page that is displayed. Click on the option button to choose the first page.

## Page Effect

The *Page Effect* option button controls the page effect that is displayed in *Full−Screen* mode. Click on the option button to select a page effect.

## Page Duration

The *Page Duration* slider controls the number of seconds that each page will be visible in *Full−Screen* mode. Drag the slider to adjust the number of seconds.

## Effect Duration

The *Effect Duration* slider controls the number of seconds that the page effect will last when changing pages. Drag the slider to adjust the number of seconds.

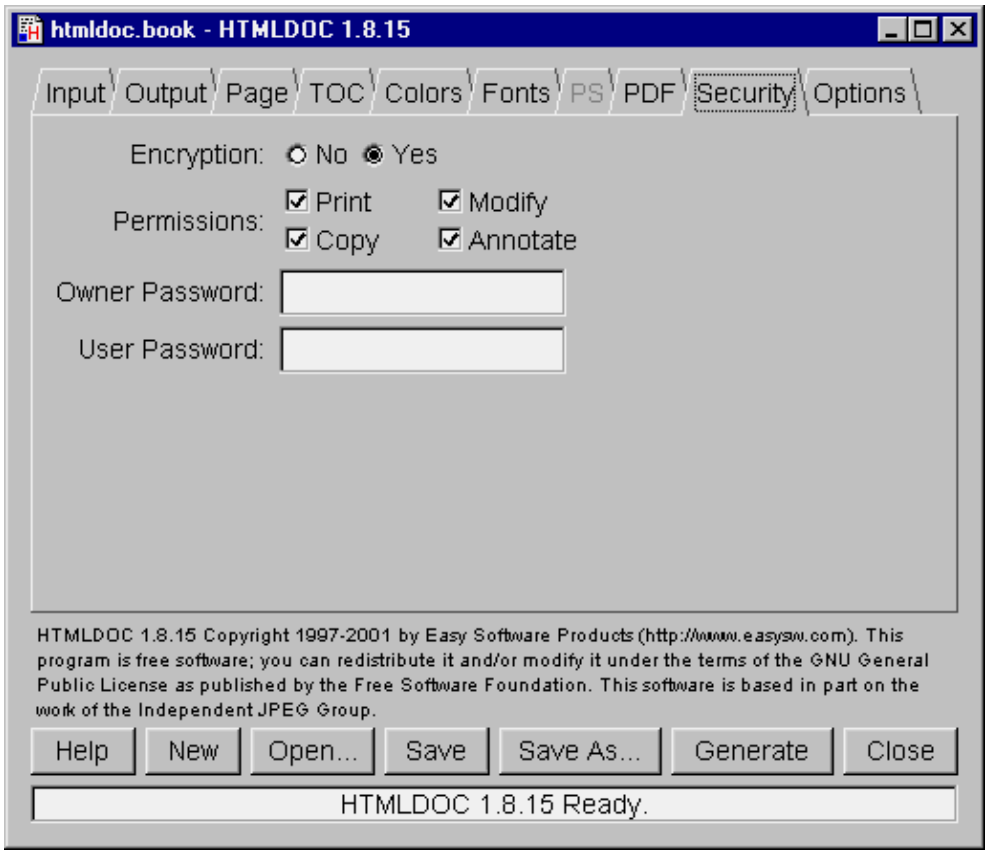



Figure 7−9 − The Security Tab

# The Security Tab

The security tab (Figure 7−9) allows you to enable PDF document encryption and security features.

## Encryption

The *Encryption* buttons control whether or not encryption is performed on the PDF file. Encrypted documents can be password protected and also provide user permissions.

## Permissions

The *Permissions* buttons control what operations are allowed by the PDF viewer.

## Owner Password

The *Owner Password* field contains the document owner password, a string that is used by Adobe Acrobat to control who can change document permissions, etc.

If this field is left blank, a random 32−character password is generated so that no one can change the document using the Adobe tools.

## User Password

The *User Password* field contains the document user password, a string that is used by Adobe Acrobat to restrict viewing permissions on the file.

If this field is left blank, any user may view the document without entering a password.

## Options

The *Include Links* and *Use TrueType Fonts* check boxes control whether or not hyperlinks and TrueType fonts are included in PDF output.

The *Include Links* option controls whether or not the internal links in a document are included in the PDF output. The document outline (shown to the left of the document in Acrobat Reader) is unaffected by this setting.

The *Use TrueType Fonts* option maps the normal Type1 fonts to TrueType fonts in the PDF file:

| Type 1 Font | TrueType Font |
|---|---|
| Courier | Courier New |
| Times | Times New Roman |
| Helvetica | Arial |
| Symbol | Symbol |

The primary purpose of this option is to allow the use of TrueType fonts that contain the full set of characters for a particular language. Most Type 1 fonts only contain the characters needed for ISO−8859−1.

## User Password

The *User Password* field contains the document user password, a string that is used by Adobe Acrobat to restrict viewing permissions on the file.

If this field is left blank, any user may view the document without entering a password.

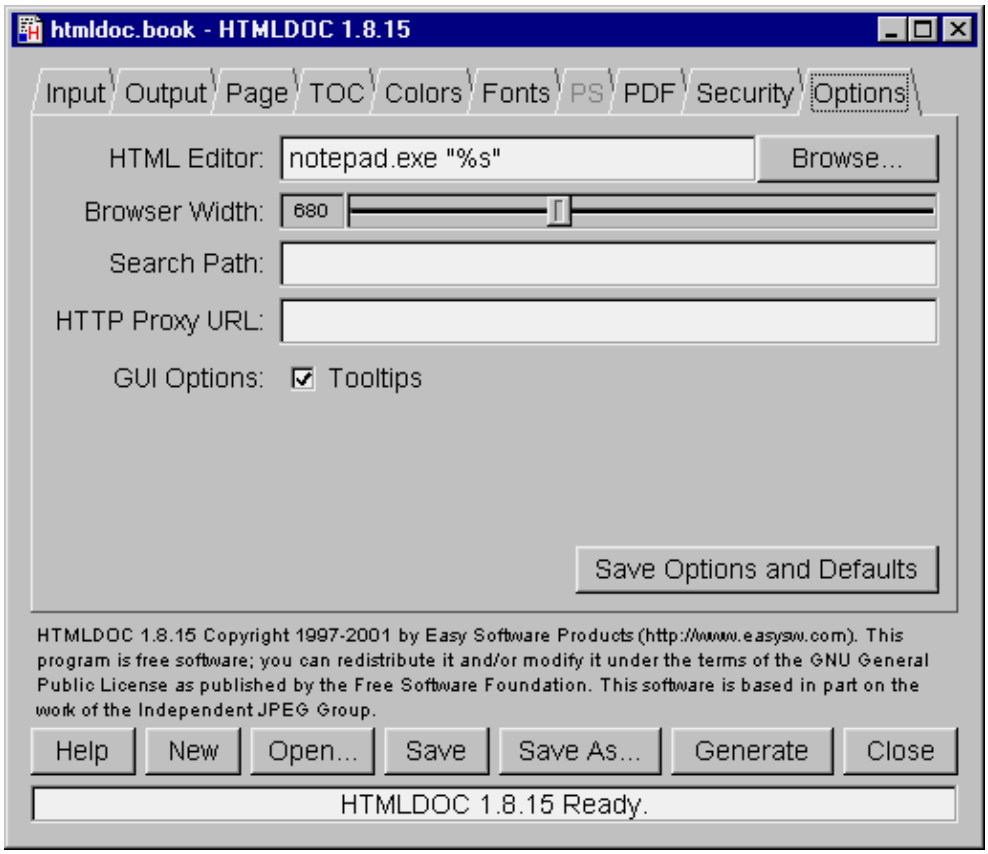



Figure 7−10 − The Options Tab

# The Options Tab

The options tab (Figure 7−10) contains the HTML file editor of your choice and allows you to save the settings and options that will be used in new documents.

## HTML Editor

The *HTML Editor* field contains the name of the HTML editor to run when you double−click on an input file or click on the *Edit Files...* button. Enter the program name in the field or click on the *Browse...* button to select the editor using the file chooser.

The `%s` is added automatically to the end of the command name to insert the name of the file to be edited. If you are using Netscape Composer to edit your HTML files you should put "−edit" before the `%s` to tell Netscape to edit the file and not display it.

## Browser Width

The *Browser Width* slider specifies the width of the browser in pixels that is used to scale images and other pixel measurements to the printable page width. You can adjust this value to more closely match the formatting on the screen.

### Search Path

The *Search Path* field specifies a search path for files that are loaded by HTMLDOC. It is usually used to get images that use absolute server paths to load.

Directories are separated by the semicolon (;) so that drive letters (and eventually URLs) can be specified.

### Proxy URL

The *Proxy URL* field specifies a URL for a HTTP proxy server.

### Tooltips

The *Tooltips* check button controls the appearance of tooltip windows over GUI controls.

### Save Options and Defaults

The *Save Options and Defaults* button saves the HTML editor and all of the document settings on the other tabs for use in new documents. These settings are also used by the command−line version of *HTMLDOC*.

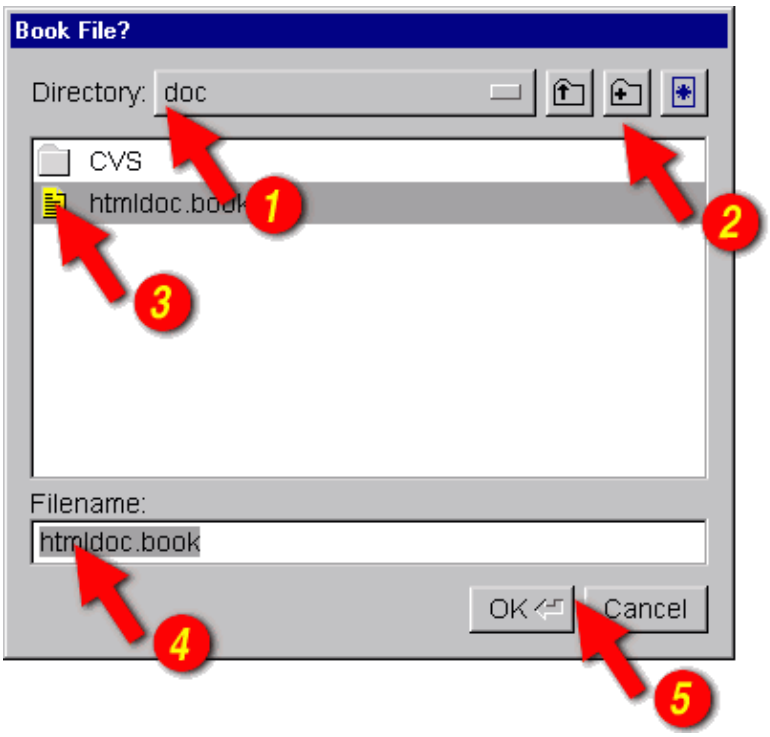



Figure 7−11 − The File Chooser

## The File Chooser

The file chooser (Figure 7−11) allows you to select one or more files and create files and directories.

## Directory

The *Directory* option button (1) shows the current directory or folder that is displayed in the file list (3). Click on the option button to navigate to other directories or folders.

## Directory Buttons

The directory buttons (2) allow you to go up one level in the directory hierarchy, create a new directory, and show all files in the directory, respectively.

## File List

The file list (3) lists the files and directories in the current directory or folder. Double−click on a file or directory to select that file or directory. Drag the mouse or hold the **CTRL** key down while clicking to select multiple files.

## Filename

The *Filename* field contains the currently selected filename. Type a name in the field to select a file or directory. As you type, any matching filenames will be highlighted; press the **TAB** key to accept the matches.

## Dialog Buttons

The dialog buttons (5) close the file chooser dialog window. Click on the *OK* button to accept your selections or the *Cancel* button to reject your selections and cancel the file operation.

# Chapter 8 − Command−Line Reference

This chapter describes all of the command−line options supported by *HTMLDOC*.

> **Note:** The free version of *HTMLDOC* for Windows does not include the command−line program.

## Basic Usage

The basic command−line usage for *HTMLDOC* is:

```
% htmldoc options filename1.html ... filenameN.html ENTER
% htmldoc options filename.book ENTER
```

The first form converts the named HTML files to the specified output format immediately. The second form loads the specified `.book` file and displays the *HTMLDOC* window, allowing a user to make changes and/or generate the document interactively.

If no output file or directory is specified, then all output is sent to the standard output file.

## Options

The following command−line options are recognized by *HTMLDOC*.

## −d directory

The −d option specifies an output directory for the document files.

This option is not compatible with the PDF output format.

## −f filename

The −f option specifies an output file for the document.

## −t format

The −t option specifies the output format for the document and can be one of the following:

| Format | Description |
|---|---|
| html | Generate one or more indexed HTML files. |
| pdf | Generate a PDF file (default version − 1.3). |
| pdf11 | Generate a PDF 1.1 file for Acrobat Reader 2.0. |
| pdf12 | Generate a PDF 1.2 file for Acrobat Reader 3.0. |
| pdf13 | Generate a PDF 1.3 file for Acrobat Reader 4.0. |
| pdf14 | Generate a PDF 1.4 file for Acrobat Reader 5.0. |
| ps | Generate one or more PostScript files (default level). |
| ps1 | Generate one or more Level 1 PostScript files. |
| ps2 | Generate one or more Level 2 PostScript files. |
| ps3 | Generate one or more Level 3 PostScript files. |

## −v

The −v option specifies that progress information should be sent/displayed to the standard error file.

## −−batch filename.book

The −−batch option specifies a book file that you would like to generate without the GUI popping up. This option can be combined with other options to generate the same book in different formats and sizes:

```
% htmldoc --batch filename.book -f filename.ps ENTER
% htmldoc --batch filename.book -f filename.pdf ENTER
```

## −−bodycolor color

The −−bodycolor option specifies the background color for all pages in the document. The color can be specified by a standard HTML color name or as a 6−digit hexadecimal number of the form #RRGGBB.

## --bodyfont typeface

The `--bodyfont` option specifies the default text font used for text in the document body. The `typeface` parameter can be one of the following:

| typeface | Actual Font |
|---|---|
| Arial | Helvetica |
| Courier | Courier |
| Helvetica | Helvetica |
| Monospace | Courier |
| Sans-Serif | Helvetica |
| Serif | Times |
| Symbol | Symbol |
| Times | Times |

## --bodyimage filename

The `--bodyimage` option specifies the background image for all pages in the document. The supported formats are GIF, JPEG, and PNG.

## --book

The `--book` option specifies that the input files comprise a book with chapters and headings.

## --bottom margin

The `--bottom` option specifies the bottom margin. The default units are points (1 point = 1/72nd inch); the suffixes "in", "cm", and "mm" specify inches, centimeters, and millimeters, respectively.

This option is only available when generating PostScript or PDF files.

## --browserwidth pixels

The `--browserwidth` option specifies the browser width in pixels. The browser width is used to scale images and pixel measurements when generating PostScript and PDF files. It does not affect the font size of text.

The default browser width is 680 pixels which corresponds roughly to a 96 DPI display.

This option is only available when generating PostScript or PDF files.

## --charset charset

The `--charset` option specifies the 8-bit character set encoding to use for the entire document. *HTMLDOC* comes with the following character set files:

| charset | Character Set |
|---|---|
| iso-8859-1 | ISO-8859-1 |
| iso-8859-2 | ISO-8859-2 |
| iso-8859-3 | ISO-8859-3 |
| iso-8859-4 | ISO-8859-4 |
| iso-8859-5 | ISO-8859-5 |
| iso-8859-6 | ISO-8859-6 |
| iso-8859-7 | ISO-8859-7 |
| iso-8859-8 | ISO-8859-8 |
| iso-8859-9 | ISO-8859-9 |
| iso-8859-14 | ISO-8859-14 |
| iso-8859-15 | ISO-8859-15 |
| koi8-r | KOI8-R |

## --color

The `--color` option specifies that color output is desired.

This option is only available when generating PostScript or PDF files.

## --compression[=level]

The `--compression` option specifies that Flate compression should be performed on the output file(s). The optional `level` parameter is a number from 1 (fastest and least amount of compression) to 9 (slowest and most amount of compression).

This option is only available when generating Level 3 PostScript or PDF files.

## --continuous

The `--continuous` option specifies that the input files comprise a web page (or site) and that no title page or table-of-contents should be generated. Unlike the `--webpage` option described later in this chapter, page breaks are not inserted between each input file.

This option is only available when generating PostScript or PDF files.

## --datadir directory

The `--datadir` option specifies the location of data files used by *HTMLDOC*.

## --duplex

The `--duplex` option specifies that the output should be formatted for two sided printing.

This option is only available when generating PostScript or PDF files. Use the `--pscommands` option to generate PostScript duplex mode commands.

## --effectduration seconds

The `--effectduration` option specifies the duration of a page transition effect in seconds.

This option is only available when generating PDF files.

## --encryption

The `--encryption` option enables encryption and security features for PDF output.

This option is only available when generating PDF files.

## --firstpage page

The `--firstpage` option specifies the first page that will be displayed in a PDF file. The `page` parameter can be one of the following:

| page | Description |
|---|---|
| p1 | The first page of the document. |
| toc | The first page of the table-of-contents. |
| c1 | The first page of chapter 1. |

This option is only available when generating PDF files.

## --fontsize size

The `--fontsize` option specifies the base font size for the entire document in points (1 point = 1/72nd inch).

## --fontspacing spacing

The `--fontspacing` option specifies the line spacing for the entire document as a multiplier of the base font size. A `spacing` value of 1 makes each line of text the same height as the font.

## --footer lcr

The `--footer` option specifies the contents of the page footer. The `lcr` parameter is a three-character string representing the left, center, and right footer fields. Each character can be one of the following:

| lcr | Description |
|---|---|
| . | A period indicates that the field should be blank. |
| : | A colon indicates that the field should contain the current and total number of pages in the chapter (n/N). |
| / | A slash indicates that the field should contain the current and total number of pages (n/N). |
| 1 | The number 1 indicates that the field should contain the current page number in decimal format (1, 2, 3, ...) |
| a | A lowercase "a" indicates that the field should contain the current page number using lowercase letters. |
| A | An uppercase "A" indicates that the field should contain the current page number using UPPERCASE letters. |
| c | A lowercase "c" indicates that the field should contain the current chapter title. |
| C | An uppercase "C" indicates that the field should contain the current chapter page number. |
| d | A lowercase "d" indicates that the field should contain the current date. |
| D | An uppercase "D" indicates that the field should contain the current date and time. |
| h | An "h" indicates that the field should contain the current heading. |
| i | A lowercase "i" indicates that the field should contain the current page number in lowercase roman numerals (i, ii, iii, ...) |
| I | An uppercase "I" indicates that the field should contain the current page number in uppercase roman numerals (I, II, III, ...) |
| l | A lowercase "l" indicates that the field should contain the logo image. |
| t | A lowercase "t" indicates that the field should contain the document title. |
| T | An uppercase "T" indicates that the field should contain the current time. |

Setting the footer to "`...`" disables the footer entirely.

## --format format

The `--format` option specifies the output format for the document and can be one of the following:

| Format | Description |
|---|---|
| html | Generate one or more indexed HTML files. |
| pdf | Generate a PDF file (default version − 1.3). |
| pdf11 | Generate a PDF 1.1 file for Acrobat Reader 2.0. |
| pdf12 | Generate a PDF 1.2 file for Acrobat Reader 3.0. |
| pdf13 | Generate a PDF 1.3 file for Acrobat Reader 4.0. |
| pdf14 | Generate a PDF 1.4 file for Acrobat Reader 5.0. |
| ps | Generate one or more PostScript files (default level). |
| ps1 | Generate one or more Level 1 PostScript files. |
| ps2 | Generate one or more Level 2 PostScript files. |
| ps3 | Generate one or more Level 3 PostScript files. |

## --gray

The `--gray` option specifies that grayscale output is desired.

This option is only available when generating PostScript or PDF files.

## --header lcr

The `--header` option specifies the contents of the page header. The `lcr` parameter is a three−character string representing the left, center, and right header fields. See the `--footer` option for the list of formatting characters.

Setting the header to "`...`" disables the header entirely.

## --headfootfont font

The `--headfootfont` option specifies the font that is used for the header and footer text. The `font` parameter can be one of the following:

- Courier
- Courier-Bold
- Courier-Oblique
- Courier-BoldOblique
- Times
- Times-Roman
- Times-Bold
- Times-Italic
- Times-BoldItalic
- Helvetica
- Helvetica-Bold
- Helvetica-Oblique
- Helvetica-BoldOblique

This option is only available when generating PostScript or PDF files.

## --headfootsize size

The `--headfootsize` option sets the size of the header and footer text in points (1 point = 1/72nd inch).

This option is only available when generating PostScript or PDF files.

## --headingfont typeface

The `--headingfont` options sets the typeface that is used for headings in the document. The `typeface` parameter can be one of the following:

| typeface | Actual Font |
|---|---|
| Arial | Helvetica |
| Courier | Courier |
| Helvetica | Helvetica |
| Monospace | Courier |
| Sans-Serif | Helvetica |
| Serif | Times |
| Symbol | Symbol |
| Times | Times |

## --help

The `--help` option displays all of the available options to the standard output file.

## --helpdir directory

The --helpdir option specifies the location of the on-line help files.

## --jpeg[=quality]

The --jpeg option enables JPEG compression of continuous-tone images. The optional quality parameter specifies the output quality from 0 (worst) to 100 (best).

This option is only available when generating Level 2 and Level 3 PostScript or PDF files.

## --landscape

The --landscape option specifies that the output should be in landscape orientation (long edge on top).

This option is only available when generating PostScript or PDF files.

## --left margin

The --left option specifies the left margin. The default units are points (1 point = 1/72nd inch); the suffixes "in", "cm", and "mm" specify inches, centimeters, and millimeters, respectively.

This option is only available when generating PostScript or PDF files.

## --linkcolor color

The --linkcolor option specifies the color of links in HTML and PDF output. The color can be specified by name or as a 6-digit hexadecimal number of the form #RRGGBB.

## --links

The --links option specifies that PDF output should contain hyperlinks.

## --linkstyle style

The --linkstyle option specifies the style of links in HTML and PDF output. The style can be "plain" for no decoration or "underline" to underline links.

## --logoimage filename

The --logoimage option specifies the logo image for the HTML navigation bar and page headers and footers for PostScript and PDF files. The supported formats are GIF, JPEG, and PNG.

## --no-compression

The --no-compression option specifies that Flate compresion should not be performed on the output files.

## --no-duplex

The `--no-duplex` option specifies that the output should be formatted for one sided printing.

This option is only available when generating PostScript or PDF files. Use the `--pscommands` option to generate PostScript duplex mode commands.

## --no-encryption

The `--no-encryption` option specifies that no encryption/security features should be enabled in PDF output.

This option is only available when generating PDF files.

## --no-jpeg

The `--no-jpeg` option specifies that JPEG compresion should not be performed on large images.

## --no-links

The `--no-links` option specifies that PDF output should not contain hyperlinks.

## --no-numbered

The `--no-numbered` option specifies that headings should not be numbered.

## --no-pscommands

The `--no-pscommands` option specifies that PostScript device commands should not be written to the output files.

## --no-title

The `--no-title` option specifies that the title page should not be generated.

## --no-toc

The `--no-toc` option specifies that the table-of-contents pages should not be generated.

## --no-truetype

The `--no-truetype` option specifies that TrueType fonts should not be used in PDF output.

## --no-xrxcomments

The `--no-xrxcomments` option specifies that Xerox PostScript job comments should not be written to the output files.

This option is only available when generating PostScript files.

## --numbered

The `--numbered` option specifies that headings should be numbered.

## --outdir directory

The `--outdir` option specifies an output directory for the document files.

This option is not compatible with the PDF output format.

## --outfile filename

The `--outfile` option specifies an output file for the document.

## --owner-password password

The `--owner-password` option specifies the owner password for a PDF file. If not specified or the empty string (""), a random password is generated.

This option is only available when generating PDF files.

## --pageduration seconds

The `--pageduration` option specifies the number of seconds that each page will be displayed in the document.

This option is only available when generating PDF files.

## --pageeffect effect

The `--pageeffect` option specifies the page effect to use in PDF files. The `effect` parameter can be one of the following:

| effect | Description |
|---|---|
| none | No effect is generated. |
| bi | Box Inward |
| bo | Box Outward |
| d | Dissolve |
| gd | Glitter Down |
| gdr | Glitter Down and Right |
| gr | Glitter Right |
| hb | Horizontal Blinds |
| hsi | Horizontal Sweet Inward |
| hso | Horizontal Sweep Outward |
| vb | Vertical Blinds |
| vsi | Vertical Sweep Inward |
| vso | Vertical Sweep Outward |
| wd | Wipe Down |
| wl | Wipe Left |
| wr | Wipe Right |
| wu | Wipe Up |

This option is only available when generating PDF files.

## --pagelayout layout

The `--pagelayout` option specifies the initial page layout in the PDF viewer. The `layout` parameter can be one of the following:

| layout | Description |
|---|---|
| single | A single page is displayed. |
| one | A single column is displayed. |
| twoleft | Two columns are displayed with the first page on the left. |
| tworight | Two columns are displayed with the first page on the right. |

This option is only available when generating PDF files.

## --pagemode mode

The `--pagemode` option specifies the initial viewing mode in the PDF viewer. The `mode` parameter can be one of the following:

| mode | Description |
|---|---|
| document | The document pages are displayed in a normal window. |
| outline | The document outline and pages are displayed. |
| fullscreen | The document pages are displayed on the entire screen in "slideshow" mode. |

This option is only available when generating PDF files.

## --path "dir1;dir2;dir3;...;dirN"

The `--path` option specifies a search path for files that are loaded by HTMLDOC. It is usually used to get images that use absolute server paths to load.

Directories are separated by the semicolon (;) so that drive letters and URLs can be specified.

## --permissions permission

The `--permissions` option specifies the document permissions. Multiple options can be specified as needed:

| Permission | Description |
|---|---|
| all | All permissions |
| annotate | User can annotate document |
| copy | User can copy text and images from document |
| modify | User can modify document |
| print | User can print document |
| no-annotate | User cannot annotate document |
| no-copy | User cannot copy text and images from document |
| no-modify | User cannot modify document |
| no-print | User cannot print document |
| none | No permissions |

This option is only available when generating PDF files.

## --portrait

The `--portrait` option specifies that the output should be in portrait orientation (short edge on top).

This option is only available when generating PostScript or PDF files.

## --pscommands

The `--pscommands` option specifies that PostScript device commands should be written to the output files.

This option is only available when generating Level 2 and Level 3 PostScript files.

## --quiet

The `--quiet` option prevents error messages from being sent to stderr.

## --right margin

The `--right` option specifies the right margin. The default units are points (1 point = 1/72nd inch); the suffixes "in", "cm", and "mm" specify inches, centimeters, and millimeters, respectively.

This option is only available when generating PostScript or PDF files.

## --size size

The `--size` option specifies the page size. The `size` parameter can be one of the following standard sizes:

| size | Description |
|---|---|
| Letter | 8.5x11in (216x279mm) |
| A4 | 8.27x11.69in (210x297mm) |
| Universal | 8.27x11in (210x279mm) |

Custom sizes are specified by the page width and length separated by the letter "x" to select a custom page size. Append the letters "in" for inches, "mm" for millimeters, or "cm" for centimeters.

This option is only available when generating PostScript or PDF files. Use the `--pscommands` option to generate PostScript page size commands.

## --textcolor color

The `--textcolor` option specifies the default text color for all pages in the document. The color can be specified by a standard HTML color name or as a 6-digit hexadecimal number of the form `#RRGGBB`.

## --textfont typeface

The `--textfont` options sets the typeface that is used for text in the document. The `typeface` parameter can be one of the following:

| typeface | Actual Font |
|---|---|
| Arial | Helvetica |
| Courier | Courier |
| Helvetica | Helvetica |
| Monospace | Courier |
| Sans-Serif | Helvetica |
| Serif | Times |
| Symbol | Symbol |
| Times | Times |

## --title

The `--title` option specifies that a title page should be generated.

## --titlefile filename

The `--titlefile` option specifies a HTML file to use for the title page.

## --titleimage filename

The `--titleimage` option specifies the title image for the title page. The supported formats are BMP, GIF, JPEG, and PNG.

## --tocfooter lcr

The `--tocfooter` option specifies the contents of the table-of-contents footer. The `lcr` parameter is a three-character string representing the left, center, and right footer fields. See the `--footer` option for the list of formatting characters.

Setting the TOC footer to "`...`" disables the TOC footer entirely.

## --tocheader lcr

The `--tocheader` option specifies the contents of the table-of-contents header. The `lcr` parameter is a three-character string representing the left, center, and right header fields. See the `--footer` option for the list of formatting characters.

Setting the TOC header to "`...`" disables the TOC header entirely.

## --toclevels levels

The `--toclevels` options specifies the number of heading levels to include in the table-of-contents pages. The `levels` parameter is a number from 1 to 6.

## --toctitle string

The `--toctitle` options specifies the string to display at the top of the table-of-contents; the default string is "Table of Contents".

## --top margin

The `--top` option specifies the top margin. The default units are points (1 point = 1/72nd inch); the suffixes "in", "cm", and "mm" specify inches, centimeters, and millimeters, respectively.

This option is only available when generating PostScript or PDF files.

## --truetype

The `--truetype` option specifies that TrueType fonts should be used in PDF output.

## --user-password password

The `--user-password` option specifies the user password for a PDF file. If not specified or the empty string (""), no password will be required to view the document.

This option is only available when generating PDF files.

## --verbose

The `-v` option specifies that progress information should be sent/displayed to the standard error file.

## --webpage

The `--webpage` option specifies that the input files comprise a web page (or site) and that no title page or table-of-contents should be generated. *HTMLDOC* will insert a page break between each input file.

This option is only available when generating PostScript or PDF files.

## --xrxcomments

The `--xrxcomments` option specifies that Xerox PostScript job comments should be written to the output files.

This option is only available when generating PostScript files.

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# Appendix B − Book File Format

This appendix describes the *HTMLDOC .book* file format.

## Introduction

The *HTMLDOC .book* file format is a simple text format that provides the command−line options and files that are part of the document. These files can be used from the GUI interface or from the command−line using the `−−batch` option:

```
htmldoc filename.book
htmldoc −−batch filename.book
```

The first form will load the book and display the GUI interface, if configured. Windows users should use *ghtmldoc.exe* executable to show the GUI and *htmldoc.exe* for the batch mode:

```
ghtmldoc.exe filename.book
htmldoc.exe −−batch filename.book
```

## The Header

Each *.book* file starts with a line reading:

```
#HTMLDOC 1.8.13
```

The version number (1.8.13) is optional.

## The Options

Following the header is a line containing the options for the book. You can use any valid command−line option on this line:

```
−f htmldoc.pdf −−titleimage htmldoc.png −−duplex −−compression=9 −−jpeg=90
```

## The Files

Following the options are a list of files or URLs to include in the document:

```
intro.html
1−install.html
2−starting.html
3−books.html
4−cmdline.html
5−cgi.html
6−htmlref.html
7−guiref.html
8−cmdref.html
a−license.html
b−book.html
c−relnotes.html
```

## Putting It All Together

The following is the complete book file needed to generate this documentation:

```
#HTMLDOC 1.8.13
−f htmldoc.pdf −−titleimage htmldoc.png −−duplex −−compression=9 −−jpeg=90
intro.html
1−install.html
2−starting.html
3−books.html
4−cmdline.html
5−cgi.html
6−htmlref.html
7−guiref.html
8−cmdref.html
a−license.html
b−book.html
c−relnotes.html
```

## Older Book Files

Prior to *HTMLDOC* version 1.8.12, the book file format was slightly different:

```
#HTMLDOC version
file count
file(s)
options
```

While *HTMLDOC* still supports reading this format, we do not recommend using it for new books. In particular, when generating a document using the `−−batch` option, some options may not be applied

correctly since the files are loaded prior to setting the output options in the old format.

# Appendix C − Release Notes

This appendix provides the release notes for each version of HTMLDOC.

## Changes in HTMLDOC v1.8.15

### New Features

- Now support media source, type, and color attributes in PS output.
- Now support per−page size, margins, headers, footers, orientation, and duplexing.
- Now support plain text for headers and footers, with $ variables to include page numbers and so forth.
- New device control prolog file for printer−specific option commands.
- Now support a new continuous web page mode that doesn't automatically insert a page break with each HTML file or URL (−−continuous).
- Now draw border around inline images as needed.
- Now support MacOS X (only command−line at present).
- Now support the "page−break−before", "text−align", "vertical−align" style attributes, but only for style information in an element's STYLE attribute.

### Changes

- Now load images into memory only as needed, and unload them when no longer needed. This provides a dramatic reduction in memory usage with files that contain a lot of in−line images.
- Now use the long names for the Flate and DCT filters in all non−inline PDF streams. This avoids a stupid bug in Acrobat Reader when printing to PostScript printers.

- HTMLDOC now strips any trailing GET query information when saving the start of files (target) in a document.
- Unqualified URLs (no leading scheme name, e.g. http:) now default to the HTTP port (80) instead of the IPP port (631).
- Optimized the image writing code to do more efficient color searching. This provides a significant speed improvement when including images.
- Now hide all text inside SCRIPT, SELECT, and TEXTAREA elements.
- OS/2 port changes from Alexander Mai.

### Bug Fixes

- If a document started with a heading greater than H1, HTMLDOC would crash.
- Full justification would incorrectly be applied to text ending with a break.
- Images using ALIGN="MIDDLE" were not centered properly on the baseline.
- Table cells that used both ROWSPAN and COLSPAN did not format properly (the colspan was lost after the first row.)
- Tables that used cells that exclusively used COLSPAN did not format properly.
- When writing HTML output, image references would incorrectly be mapped using the current path.
- Images with a width or height of 0 should not be written to PS or PDF output.
- The CreationDate comment in PostScript output contained a bad timezone offset (+−0500, for example, instead of −0500).
- The PHP portal example now verifies that the URL passed to it contains no illegal characters.

## Changes in HTMLDOC v1.8.14

### New Features

- Added support for 128−bit encryption.
- Added support for GET form request data in the PHP and Java "portal" examples.

### Changes

- Most output generation limits have been removed; HTMLDOC now dynamically allocates memory as needed for pages, images, headings, and links. This has the happy side−effect of reducing the initial memory footprint significantly.
- Now call setlocale() when it is available to localize the date and time in the output.
- The table parsing code now checks to see that a ROWSPAN attribute fits in the table; e.g., a ROWSPAN of 10 for a table that has only 6 rows remaining needs to be reduced to 6...

### Bug Fixes

- Tables with a lot of COLSPANs could cause a divide− by−zero error or bad pages (NAN instead of a number.)
- Table cells with a single render element would not be vertically aligned.
- The −−quiet option would enable progress messages on the command−line.
- Table cell widths could be computed incorrectly, causing unnecessary wrapping.

# Changes in HTMLDOC v1.8.13

## New Features

- Added support for secure (https) URLs via the OpenSSL library.
- Added support for Acrobat 5.0 (PDF 1.4).
- Added support for transparency in PostScript and PDF 1.1 and 1.2 output.
- Added a −−no−jpeg option (same as −−jpeg=0)
- Added support for the CSS2 page−break−before and page−break−after properties.
- Added a PHP example.

## Changes

- External file references to non−PDF files now use the "Launch" action so they can be opened/executed/saved as allowed by the OS and PDF viewer.
- Changed the indexed/JPEG'd transition point to 256 colors when using Flate compression. This makes PDF files much smaller in general.
- Changed the in−line image size limit to 64k.
- Now allow "<" followed by whitespace, "=", or "<". This violates the HTML specification, but we're sick of people complaining about it.
- Preferences are now stored in a user−specific file under Windows, just like UNIX. This provides user−specific preferences and allows preferences to be kept when upgrading to new versions of HTMLDOC.
- The book loading code now allows for blank lines, even though these are not a part of the format. (added to support some scripted apps that include extra newlines...)
- Changed the leading space handling of blocks to more closely match the standard browser behavior.

## Bug Fixes

- The table formatting code adding the border width to the cell width, while Netscape and MSIE don't. This caused some interesting formatting glitches...
- The table formatting code didn't account for the preferred width of colspan'd cells.
- The table formatting code tried to enforce the minimum cell width when squeezing a table to fit on the page; this caused the table to still exceed the width of the page.
- The PDF catalog object could contain a reference to a /Names object of "0 0 R", which is invalid. This would happen when the "−−no−links" option was used.
- Several HTML elements were incorrectly written with closing tags.
- When piping PDF output, the temporary file that is created needed to be open for reading and writing, but HTMLDOC only opened the file for writing.
- Image links did not work.
- The JPEG image loading code did not correctly handle grayscale JPEG images.
- JPEG images were not encrypted when writing a document with encryption enabled.
- The user password was not properly encrypted.
- The colormap of indexed images were not encrypted when writing a document with encryption enabled.
- The temporary file creation and cleanup functions did not use the same template under Windows, causing multiple conversions to fail when temporary files were used.
- Paragraphs could end up with one extra text fragment, causing the line to be too long.
- The command−line program would clear the error count after reading all the files/URLs on the command−line, but before generating the document. If there were problems reading the files/URLs,

HTMLDOC would return a 0 exit status instead of 1.
- Image objects that were both JPEG and Flate compressed would not display (filters specified in the wrong order.)
- Images with more than 256 colors would cause a segfault on some systems.
- Background images would generate the error message "XObject 'Innn' is unknown".

# Changes in HTMLDOC v1.8.12

## New Features

- Added new "−−batch" option to convert HTMLDOC book files from the command−line.
- Added support for the "−display" option on systems that use X11.
- Now use image objects in PDF output for images when the image width * height * depth > 32k.
- Now use JPEG compression when the number of colors would be > 32 colors or 16 gray shades.
- True transparency support for GIF files in PDF 1.3 output!
- The GUI now automatically changes the extension of the output filename as needed.
- The GUI now collects all error messages and shows them once after the document is generated.
- Added support for HSPACE and VSPACE attributes for images with ALIGN="LEFT" or ALIGN="RIGHT".
- Added new Java interface to HTMLDOC.

## Changes

- Consolidated temporary file management into new file_temp() function. The new function also makes use of the Windows "short lived" open option which may improve performance with small temporary files.
- Updated book file format and added an appendix describing the format.
- Now default to PDF 1.3 (Acrobat 4.0) output format.
- Now output length of PDF streams with the stream object; this offers a modest reduction in file size.
- The HTTP file cache now keeps track of previous URLs that were downloaded.
- The HTTP code now supports redirections (status codes 301 to 303) to alternate URLs.
- Limit the height check for table rows to 1/8th of the page length; this seems to provide fairly consistent wrapping of tables without leaving huge expanses of blank space at the bottom of pages.
- The HTML output now also includes a font−family style for PRE text; otherwise the body font would override the PRE font with some browsers.
- The snprintf/vsnprintf emulation functions were not included in the HTMLDOC makefile.
- RGB hex colors are now recognized with or without the leading #. This breaks HTML standards compliance but should reduce the number of problem reports from buggy HTML.
- The stylesheet generated with the HTML output no longer contains absolute font sizes, just the typefaces and a relative size for SUB/SUP.
- The title image is no longer scaled to 100% in the HTML output.

## Bug Fixes

- The web page output was not divided into chapters for each input file.
- The "make install" target did a clean.
- The configure script would remove the image libraries if you did not have FLTK installed.
- The fix_filename() function didn't handle relative URLs for images (e.g. SRC="../images/filename.gif")
- Comments in the source document were being closed by a ".

- The command−line and GUI interfaces looked for "outlines" instead of "outline" for the page mode.
- The HTML output code didn't output closing tags for empty elements.
- The GUI interface started with the compression slider enabled, even for HTML output.
- The beginnings of some lines could start with whitespace.
- Wasn't aligning images and text on lines based on the line height.
- The compression slider was enabled in the GUI even though HTML output was selected.
- The Perl example code was incorrect.
- Fixed the check for whether or not pages were generated.
- htmlSetCharSet() wasn't reloading the character set data if the data directory changed.
- The GUI did not reset the default background color.
- The 'C' page number style (chapter page numbers) started at 3 instead of 1.
- The chapter links were off by 1 or 2 pages when no title page was included.

# Changes in HTMLDOC v1.8.8

## New Features

- Added support for PDF security/encryption!
- Now support TABLE height attribute.
- Now generate an error message if no pages are generated (with a suggestion to use the webpage option.)
- New "paths" option to specify additional directories to search for files. This is useful when the source files use absolute server paths.

## Changes

- Added missing casts in htmllib.cxx that were causing a compile warning with some compilers.
- No longer draw borders around empty cells in tables..
- Now disable the TOC tab when using webpage mode.
- Now scale title image to 100% in HTML output.
- Now handle comments with missing whitespace after the "<!−−".

## Bug Fixes

- Nested tables didn't take into account the table border width, spacing, or padding values.
- HTMLDOC crashed under Solaris when reading HTML files from the standard input.
- text text was rendered without an intervening space.

# Changes in HTMLDOC v1.8.7

## New Features

- The configure script now uses the local PNG, ZLIB, and/or JPEG libraries when they are new enough.
- The configure script now uses the −fno−rtti, −fno−exceptions, and −fpermissive options as needed with GCC (smaller, faster executables, works around X header bugs in Solaris.)
- Added a −−toctitle option to set the table−of−contents title from the command−line (was only available in the GUI in previous releases...)

- New "<!−− NEED amount −−>" comment to force a page feed if there is not sufficient room on the page for the following text.
- Page comments are now supported in tables.
- Table rows are now allocated dynamically, MAX_ROWS at a time.

### Changes

- Increased default MAX_PAGES to 10000 (was 5000.)
- File links in book files now point to the top of the next page.
- <TABLE ALIGN=xyz> now aligns the table (previously it just set the default alignment of cells.)
- Transparent GIFs now use the body color instead of white for the transparent color.
- Updated to LIBPNG 1.0.6 in source distribution.
- Updated the default cellpadding to be 1 pixel to match Netscape output.
- Updated line and block spacing to match Netscape.
- DL/DT/DD output now matches browsers (was indented from browser output.)
- Now only output link (A) style if it is set to "none". Otherwise Netscape would underline all targets as well as links.
- Increased the MAX_COLUMNS constant to 200, and dropped MAX_ROWS to 200. Note that the new table code now allocates rows in increments of MAX_ROWS rows, so the actual maximum number of rows depends on available memory.

### Bug Fixes

- Now ignore illegal HTML in tables.
- The VALIGN code didn't handle empty cells properly.
- Wasn't offsetting the start of each row by the cell padding.
- The JPEG image loading code didn't work for some JPEG images, particularly those from digital cameras (JPEG but not JFIF format.)
- The strikethrough line was not being drawn in the correct position.
- Wasn't setting the height of BR elements, so <BR><BR> didn't insert a blank line.
- The table of contents would show the wrong page numbers if no title page was generated.
- Cell widths did not subtract any border, padding, or spacing from the "preferred" width, causing formatting differences between web browsers and HTMLDOC.
- The PNG loading code did not handle interlacing or transparency.
- The HTML parsing code did not prevent elements in embedded files from completing elements in the parent file.
- The table CELLSPACING amount was being applied twice in the table sizing calculations.

## Changes in HTMLDOC v1.8.6

### New Features

- New linkcolor and linkstyle options.

### Changes

- Minor source changes for OS/2 compilation.
- SUP and SUB now raise/lower text more to be consistent with browser look−n−feel.
- Non−breaking space by itself was being output. Now check for that and ignore strings that consist

entirely of whitespace.
- New progress bar.

### Bug Fixes

- Didn't add whitespace after a table caption.
- Nested tables caused formatting problems (flatten_tree() didn't insert breaks for new rows)
- A cell whose minimum width exceeded the available width for the table would cause the table to go off the page.
- Cells that spanned more than two pages were drawn with boxes around them rather than just the sides.
- The stylesheet info in the HTML output specified the H1 size for all headings.
- The title page was incorrectly formatted when an image was specified − the text start position was computed using the pixel height of the title image and not the formatted height.
- 1 color images didn't come out right; the "fix" to work around an Acrobat Reader bug was being done too soon, so the color lookups were wrong.
- HTML file links now work properly.
- Now limit all HTML input to the maximum size of input buffers to avoid potential buffer overflow problems in CGIs.
- If a row had a predefined height, HTMLDOC wasn't making sure that the row would fit on the current page.
- THEAD, TFOOT, and TBODY caused problems when formatting tables. Note: THEAD and TFOOT are *still* not supported, however the code now properly ignores them and parses the rows in the TBODY group.
- The VALIGN code introduced in the 1.8.5 release didn't check for NULL pointers in all cases.

## Changes in HTMLDOC v1.8.5

### New Features

- New "−−titlefile" option to include an HTML file for the title page(s).
- New 'C' header/footer option to show current page number within chapter or HTML file.
- Allow adding of .book files to import all HTML files in the book.
- New "HALF PAGE" page comment to feed 1/2 page.
- Added VALIGN and HEIGHT support in tables.

### Changes

- Now optimize link objects in PDF files (provides a 40k reduction in file size for the HTMLDOC manual alone)
- Table rows that cross page boundaries are now rendered more like Netscape and MSIE.
- Now support HTMLDOC_DATA and HTMLDOC_HELP environment variables under UNIX (for alternate install directory)
- Now show error messages when HTMLDOC can't open the AFM, character set, or PostScript glyph files.
- The logo image is now scaled to its "natural" size (as it would appear in a web browser)
- Now recognize VALIGN="MIDDLE" or VALIGN="CENTER".

### Bug Fixes

- Generation of PDF files to the standard output (i.e. to the web server + browser) didn't work on some versions of UNIX. HTMLDOC now writes the PDF output to a temporary file and then copies it to the standard output as needed.
- PDF links were missing the first 5 characters in the filename; the code was trying to skip over the "file:" prefix, but that prefix was already skipped elsewhere.
- Nested descriptive lists (DL) did not get rendered properly.
- Tables had extra whitespace before and after them.
- Multiple aligned images confused parse_paragraph(); the images would overlap instead of stack on the sides.

## Changes in HTMLDOC v1.8.4

### Changes

- More configure script changes for FLTK DSOs.
- FileIcon.cxx was still using NULL for outline (an integer), which caused some ANSI C++ compilers to complain.

### Bug Fixes

- The Fonts and Colors tab groups did not extend to the full width of the tab area, which prevented the Browse button from working when clicked on the right side.
- The help dialog window did not scroll all the way to the bottom of the text.
- The chapter ("c") header/footer string did not work.
- The heading ("h") header/footer string did not always match the first heading on a page.
- The header and footer fonts were not used when computing the widths of the header and footer strings.
- The Windows distribution did not create the right shortcut for the Users Manual in the Start menu.
- The command−line code did not accept "−−grayscale", only "−−gray"
- Multi−file HTML output did not use the right link for the table−of−contents file if no title page was being generated.
- Extra whitespace before and after tables has been eliminated.

## Changes in HTMLDOC v1.8.3

### New Features

- New "−−browserwidth" option to control scaling of images and tables that use fixed pixel widths.

### Changes

- The configure script now looks for the OpenGL library (required if you use a shared FLTK library with OpenGL support.)
- Increased the max number of chapters to 1000.

### Bug Fixes

- Page break comments didn't force a paragraph break.
- −−no−toc prevented chapters from being output in PS and PDF files.
- Filenames didn't always get updated properly when doing a "save as"...
- Fixed some more leading/trailing whitespace problems.
- Wasn't freeing page headings after the document was generated.
- Wasn't range checking the current chapter number; now limits the number of chapters to MAX_CHAPTERS and issues an error message whenever the limit is exceeded.

## Changes in HTMLDOC v1.8.2

### New Features

- New "setup" program for UNIX software installation.

### Changes

- Documentation updated for new UNIX "setup" program and "..." usage for headers and footers.
- Changed margins to floating point (instead of integer) to improve table column accuracy.

### Bug Fixes

- HTMLDOC could crash under Microsoft Windows with some types of HTML files. This was caused by a stack overflow, usually when processing nested tables.
- Multiple HTML files weren't being converted properly in web page mode − only the last file would be generated for PostScript output, and no file for PDF output.
- Wasn't preserving the whitespace between "one" and "two" in the HTML code "one<I> two</I> three".
- Paragraph spacing was inconsistent.
- <TABLE WIDTH="xx"> wasn't formatted properly.
- The command−line code wasn't opening HTML files in binary mode. This caused problems under Microsoft Windows.

## Changes in HTMLDOC v1.8.1

### Changes

- The configure script didn't update the ARFLAGS variable for *BSD operating systems (no "s" option to build the symbol table...)
- Changed the installation commands to only create the installation directory if it does not exist. This prevents installation errors on some platforms the second time around.
- Now use the Microsoft definitions for characters 128 through 159 that are otherwise unused by the ISO−8859−x character sets.
- Now set optimization settings when we know the compiler.
- Now always quote attribute values in HTML output to make HTML lint programs happy.

### Bug Fixes

- Wasn't using TOC title string in PDF document outline.
- Preformatted text in tables didn't force the column width.
- Cells using COLSPAN > 1 didn't contribute to the width of columns.
- The table code didn't enforce the per−column minimums under certain circumstances, causing "scrambled" columns.
- The configure script and makefiles didn't work when FLTK was not available. They now only build the "gui" library when it is available.
- The Windows distribution was installing files under PROGRAMDIR instead of TARGETDIR. This prevented users from customizing the installation directory.
- The configure script overrode the LDFLAGS environment variable, preventing FLTK from being located in a non− default directory.

## Changes in HTMLDOC v1.8

### New Features

- Now support PDF 1.1 (Acrobat 2.x) and PDF 1.3 (Acrobat 4.0).
- Now support PDF page modes, layouts, and effects, and the first page that is displayed in Acrobat Reader.
- Now support PostScript Level 3 output with Flate image compression.
- Now support PostScript commands for page size and duplexing.
- Now add filenames as needed to HTML links.
- Added optimizations to output code to further reduce PDF and PostScript file size.
- Now support alternate 8−bit character sets. Currently we supply data files for the ISO−8859−N character sets.
- Added chapter headings to the available header/footer formats.
- The GUI file chooser is significantly improved and supports selection of multiple HTML files.
- The GUI now provides on−line help.
- Many other GUI improvements.
- Added support for DIR and MENU block elements.
- The header and footer text can now be made boldface, italic, etc.
- Font settings are now exported to HTML files in a style sheet.
- Now support page breaks using HTML comments.
- The image dimensions are now exported to HTML files.
- Added landscape printing option.
- Added CAPTION support for tables.
- Filename links now work for HTML files included in a document.
- Now support BGCOLOR in tables.

### Changes

- Lots of documentation changes.
- Much better table formatting.
- Changed HTML output to use less invasive navigation bars at the top and bottom of each file. This also means that the "−−barcolor" option is no longer supported!
- Updated to use existing filenames in HTML (directory) output.
- Now recognize any local PDF file as a local file link (i.e. you just need "HREF=filename.pdf" and not "HREF=file:filename.pdf")

- <TT>, <CODE>, and <SAMP> no longer reduce the font size.
- Now put whitespace after image data in PDF files. This change was needed to work around a bug in Acrobat Reader 4.0.
- Now generate a complete encoding vector for fonts in PDF files. This change was needed to work around a bug in all versions of Acrobat Exchange that did not recognize the WinANSI encoding defined in the PDF specifications.
- Now filter out the BREAK attribute from HR elements.
- Now only load images once.

### Bug Fixes

- Wasn't escaping &,<, or > in HTML output
- Wasn't preserving  
- Links in multi−file HTML output were off−by−one.
- BLOCKQUOTE needed to be like CENTER and DIV.
- Needed to use existing link name if present for headings to avoid nested link name bug in Netscape and MSIE.
- Extremely long link names could cause TOC generation to fail and HTMLDOC to crash.
- PDF output was not compatible with Ghostscript/Ghostview because Ghostscript does not support inherited page resources or the "Fl" abbreviation for the "FlateDecode" compression filter.
- PostScript DSC comments didn't have unique page numbers. This caused Ghostview (among others) to get confused.
- Some functions didn't handle empty text fragments.
- Images couldn't be scaled both horizontally and vertically.
- didn't support the VALUE attribute (but
  did...)
  1. Fixed whitespace problems before and after some markups that was caused by intervening links.
  2. The indexed image output code could generate an image with only 1 color index used, which upset Acrobat Reader.
  3. Fixed a bug in table−of−contents handling
  4. HTMLDOC would crash on some systems if you converted a web page on the command−line.
  5. Wasn't setting the font size and spacing soon enough when generating files on the command−line.
  6. Didn't hide EMBED elements when generating indexed HTML files.
  7. Didn't always set the current drawing position before drawing a box or line.
  8. Base85 encoding of image data was broken for PostScript output.
  9. JPEG compression was broken for PostScript output.
  10. Didn't set binary mode for the standard output under Windows and OS/2 needed.

## Changes in HTMLDOC v1.7

- Added option for overriding the background color or image.
- Added default font typeface and size options.
- Added progress indicator for page formatting.
- The HTMLDOC window is now resizeable.
- The <TABLE> and <CENTER> markups didn't start a new block.
- strcasecmp and friends are not available on all platforms.
- Added support for MacOS (command−line only).

- The width of table cells could be off by 1 point causing unnecessary text wrapping.
- The GUI's default center footer wasn't "blank".
- Images could be "lost" if they reside in the current directory or use an absolute path.
- Documents without titles or headings could crash HTMLDOC.
- The image loading code could crash due to a MSVC++ runtime library bug.
- Spacing before <HR>'s wasn't consistent.
- Buffer overflow problems causing crashes.
- Didn't accept whitespace in variables, e.g. "<TAG NAME = VALUE>"
- Links didn't always get propagated.
- The Flate compressor data was not getting freed, so HTMLDOC could use a lot of memory when compression was enabled.

## Changes in HTMLDOC v1.6

- Now support JPEG compression of images.
- Now have selectable Flate (ZIP) compression level.
- Now only adjust top and bottom margins if headers and footers are used.
- Better HTML output support (now remember files for links in multi−file output).
- Increased maximum page count to 5000.
- Needed to show headers on all pages in web page mode.
- Now recognize both "in" and "inch" for measurements.
- <BR> was not handled properly.
- Selecting "web page" in the GUI clears the title toggle.
- TABLE row spacing was not right...
- <TD COLSPAN=n> now draws multi−column borders.
- Column widths were computed wrong when COLSPAN was used.
- Nested lists were not handled right.
- Internal links didn't work for PDF output.
- Block spacing should now be more consistent.
- Image scaling was off
- now only use page width so that images are not warped.
- The footer was always one line too low.
- Couldn't double−click on input filename to edit.

## Changes in HTMLDOC v1.5

- Added customization of headers and footers.
- Added new "−−title" image option.
- Can now put logo image in header or footer.
- <MARKUP ID="name"> now works for link destinations.
- The table of contents now appears as part of the document outline in PDF output.
- Links to local PDF files are now treated as file links in PDF output instead of web links.
- You can now turn the title page on/off as desired.
- PostScript and PDF output to stdout now works.
- Nested tables now format properly.
- <HR> now provides horizontal rule; to get a page break use "<HR BREAK>".
- Fixed <TABLE BORDER=0> bug.
- Fixed GIF loader bug (caused problems on Alpha machines)
- No longer get extra line after list items.
- <FONT> markup nesting now works.

- ""by itself would cause loss of 15 characters.
- The current directory was not tracked properly under Windows.
- The right, top, and bottom margins were not being saved properly.
- The htmlReadFile() function could consume too much stack space, causing a program failure.
- PostScript and PDF files were corrupt when generating a web page with a title page.

## Changes in HTMLDOC v1.4

- Now use autoconf "configure" script to build UNIX makefile.
- Now handle relative filenames a lot better when loading images and files.
- Added "−−webpage" option to support printing of plain HTML files (i.e. not documents with chapters)
- Added support for document backgrounds in PostScript and PDF output
- Added "−−no−toc" and "−−no−title" options to disable the table−of−contents and title pages, respectively
- PDF files now store all named links for use from a web page (HREF="filename.pdf#name")
- Converted to C++
- Now using FLTK for the GUI under UNIX and Windows (yeah, one set of code!)
- Merged GUI and command−line versions
- Greatly enhanced GUI now supports nearly all command−line options.
- Miscellaneous fixes to HTML parsing code
- PDF links should now go to the right page all the time
- Fixed DSC comments in PostScript output to conform to the standard
- Fixed dumb bug in Windows version
- didn't handle HTML files with only a LF at the end of each line (this is a BUG in the MSVC++ runtime libraries!)
- <PRE> inside a list didn't work
- parse_table() and friends didn't check for a NULL parent pointer.
- Paragraph text that wasn't enclosed by P markups was located on the wrong page when followed by a H1 markup.

## Changes in HTMLDOC v1.3.1

- Fixed font encoding vector in PostScript output (minus instead of hyphen for '−' character).

## Changes in HTMLDOC v1.3

- New GUI for managing documents (Windows + X11/Motif)
- Better table printing with support for user−specified column widths and better automatic−sizing
- PNG loading now works when grayscale output is requested
- No image optimization was performed in PDF or Level 2 PostScript files. HTMLDOC now converts images to indexed (1,2,4,8 bits) if there is an advantage (fewer bits per pixel) and no loss of color would occur
- The filenames in links were getting lost when writing indexed HTML to a directory
- The logo image filename wasn't being localized when writing indexed HTML to a directory
- Fonts, images, and links weren't supported inside a PRE tag
- Added support for the <!DOCTYPE> markup
- No longer assume that chars are unsigned by default
- Invalid or missing links no longer generate bad PDF files

- External links (http:, ftp:, etc) now work
- Escaped characters are now decoded correctly in the table of contents in PDF files
- Image scaling is now more intelligent

## Changes in HTMLDOC v1.2

- Now support "internal" links in a document (PDF HTML).
- Added "no compression" option for PDF files; this is needed for older PDF readers like Acroread 2.x.
- Much better parsing of HTML; should now work very well with the HTML output by Netscape Composer.
- Wasn't opening image files in "binary" mode (Windows).
- The htmlReadString() and htmlWriteString() functions were removed because of portability problems to HP−UX and Windows, among others.

## Changes in HTMLDOC v1.1

- Ordered (numbered) lists are now supported, as are the TYPE=, START=, and VALUE= option variables.
- Now support coverpages for PS and PDF output with optional logo image.
- Running headings (at the bottom of PS/PDF pages) are now tracked correctly.
- Fixed parsing of lists so lists generated by Netscape Composer work right...
- Fixed HTML links when generating a single HTML file.
- The −−numbered option didn't number all headings (only those in the table−of−contents).

## Changes in HTMLDOC v1.0

- Initial version.